Namespace LlamaLogic.Packages

Classes

ArrayBufferWriterOfByteStream
: Represents a write-only Stream that collects what is written to it in memory (🔓)

DataBasePackedFile
: Represents a Maxis DBPF format package file (🔒)

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: Represents the unique key of a resource within a DataBasePackedFile

Enums

CompressionMode
: Represents the compression mode to use when setting the content of a resource in a DataBasePackedFile

ResourceFileType
: The file type of a package resource independent of storage within a DataBasePackedFile

ResourceKeyOrder
: Specifies the order in which a ResourceKey sequence should be processed

ResourceType
: A type of resource within a DataBasePackedFile

Class ArrayBufferWriterOfByteStream

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents a write-only Stream that collects what is written to it in memory (🔓)

```
public sealed class ArrayBufferWriterOfByteStream : Stream, IAsyncDisposable, IDisposable
```

Inheritance

object ← MarshalByRefObject ← Stream ← ArrayBufferWriterOfByteStream

Implements

IAsyncDisposable, IDisposable

Inherited Members

Stream.Null , Stream.BeginRead(byte[], int, int, AsyncCallback, object) , Stream.BeginWrite(byte[], int, int, AsyncCallback, object) , Stream.Close() , Stream.CopyTo(Stream) , Stream.CopyTo(Stream, int) , Stream.CopyToAsync(Stream) , Stream.CopyToAsync(Stream, int) , Stream.CopyToAsync(Stream, int, CancellationToken) , Stream.CopyToAsync(Stream, CancellationToken) , Stream.Dispose() , Stream.DisposeAsync() , Stream.EndRead(IAsyncResult) , Stream.EndWrite(IAsyncResult) , Stream.FlushAsync() , Stream.FlushAsync(CancellationToken) , Stream.Read(Span<byte>) , Stream.ReadAsync(byte[], int, int) , Stream.ReadAsync(byte[], int, int, CancellationToken) , Stream.ReadAsync(Memory<byte>, CancellationToken) , Stream.ReadByte() , Stream.Synchronized(Stream) , Stream.Write(ReadOnlySpan<byte>) , Stream.WriteAsync(byte[], int, int) , Stream.WriteAsync(byte[], int, int, CancellationToken) , Stream.WriteAsync(ReadOnlyMemory<byte>, CancellationToken) , Stream.WriteByte(byte) , Stream.CanTimeout , Stream.ReadTimeout , Stream.WriteTimeout , MarshalByRefObject.GetLifetimeService() , MarshalByRefObject.InitializeLifetimeService() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

SetLength(long) changes the length the stream reports: WrittenMemory and WrittenSpan are then truncated to it, or extended to it with zeros.

Constructors

ArrayBufferWriterOfByteStream(int?)

Represents a write-only Stream that collects what is written to it in memory ( )

```
public ArrayBufferWriterOfByteStream(int? initialCapacity = null)
```

Parameters

`initialCapacity` int?

The number of bytes to allocate initially, or null for the default

Remarks

SetLength(long) changes the length the stream reports: WrittenMemory and WrittenSpan are then truncated to it, or extended to it with zeros.

Properties

CanRead

When overridden in a derived class, gets a value indicating whether the current stream supports reading.

```
public override bool CanRead { get; }
```

Property Value

bool

true if the stream supports reading; otherwise, false.

CanSeek

When overridden in a derived class, gets a value indicating whether the current stream supports seeking.

```
public override bool CanSeek { get; }
```

Property Value

bool

true if the stream supports seeking; otherwise, false.

CanWrite

When overridden in a derived class, gets a value indicating whether the current stream supports writing.

```
public override bool CanWrite { get; }
```

Property Value

bool

true if the stream supports writing; otherwise, false.

Length

When overridden in a derived class, gets the length in bytes of the stream.

```
public override long Length { get; }
```

Property Value

long

A long value representing the length of the stream in bytes.

Exceptions

NotSupportedException

A class derived from `Stream` does not support seeking.

ObjectDisposedException

Methods were called after the stream was closed.

Position

When overridden in a derived class, gets or sets the position within the current stream.

```
public override long Position { get; set; }
```

Property Value

long

The current position within the stream.

Exceptions

IOException

An I/O error occurs.

NotSupportedException

The stream does not support seeking.

ObjectDisposedException

Methods were called after the stream was closed.

WrittenMemory

Gets the bytes written to the stream

```
public ReadOnlyMemory<byte> WrittenMemory { get; }
```

Property Value

ReadOnlyMemory<byte>

WrittenSpan

Gets the bytes written to the stream

```
public ReadOnlySpan<byte> WrittenSpan { get; }
```

Property Value

ReadOnlySpan<byte>

Methods

Flush()

When overridden in a derived class, clears all buffers for this stream and causes any buffered data to be written to the underlying device.

```
public override void Flush()
```

Exceptions

IOException

An I/O error occurs.

Read(byte[], int, int)

When overridden in a derived class, reads a sequence of bytes from the current stream and advances the position within the stream by the number of bytes read.

```
public override int Read(byte[] buffer, int offset, int count)
```

Parameters

`buffer` byte[]

An array of bytes. When this method returns, the buffer contains the specified byte array with the values between `offset` and (`offset` + `count` - 1) replaced by the bytes read from the current source.

`offset` int

The zero-based byte offset in `buffer` at which to begin storing the data read from the current stream.

`count` int

The maximum number of bytes to be read from the current stream.

Returns

int

The total number of bytes read into the buffer. This can be less than the number of bytes requested if that many bytes are not currently available, or zero (0) if the end of the stream has been reached.

Exceptions

ArgumentException

The sum of `offset` and `count` is larger than the buffer length.

ArgumentNullException

`buffer` is null.

ArgumentOutOfRangeException

`offset` or `count` is negative.

IOException

An I/O error occurs.

NotSupportedException

The stream does not support reading.

ObjectDisposedException

Methods were called after the stream was closed.

Seek(long, SeekOrigin)

When overridden in a derived class, sets the position within the current stream.

```
public override long Seek(long offset, SeekOrigin origin)
```

Parameters

`offset` long

A byte offset relative to the `origin` parameter.

`origin` SeekOrigin

A value of type SeekOrigin indicating the reference point used to obtain the new position.

Returns

long

The new position within the current stream.

Exceptions

IOException

An I/O error occurs.

NotSupportedException

The stream does not support seeking, such as if the stream is constructed from a pipe or console output.

ObjectDisposedException

Methods were called after the stream was closed.

SetLength(long)

When overridden in a derived class, sets the length of the current stream.

```
public override void SetLength(long value)
```

Parameters

`value` long

The desired length of the current stream in bytes.

Exceptions

IOException

An I/O error occurs.

NotSupportedException

The stream does not support both writing and seeking, such as if the stream is constructed from a pipe or console output.

ObjectDisposedException

Methods were called after the stream was closed.

Write(byte[], int, int)

When overridden in a derived class, writes a sequence of bytes to the current stream and advances the current position within this stream by the number of bytes written.

```
public override void Write(byte[] buffer, int offset, int count)
```

Parameters

`buffer` byte[]

An array of bytes. This method copies `count` bytes from `buffer` to the current stream.

`offset` int

The zero-based byte offset in `buffer` at which to begin copying bytes to the current stream.

`count` int

The number of bytes to be written to the current stream.

Exceptions

ArgumentException

The sum of `offset` and `count` is greater than the buffer length.

ArgumentNullException

`buffer` is null.

ArgumentOutOfRangeException

`offset` or `count` is negative.

IOException

An I/O error occurred, such as the specified file cannot be found.

NotSupportedException

The stream does not support writing.

ObjectDisposedException

Write(byte[], int, int) was called after the stream was closed.

Enum CompressionMode

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents the compression mode to use when setting the content of a resource in a DataBasePackedFile

```
public enum CompressionMode
```

Fields

`Auto = 0`

The library will choose whether and how to compress the data or not

`CallerSuppliedInternal = 6`

⚠️ The data will not be processed for compression by the library and the resource will be marked as having been compressed using Maxis' internal method; accepted by SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) and SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) only, which also take the decompressed size (use with caution)

`CallerSuppliedStreamable = 7`

⚠️ The data will not be processed for compression by the library and the resource will be marked as having been compressed using Maxis' streamable method; accepted by SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) and SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) only, which also take the decompressed size (use with caution)

`CallerSuppliedZLib = 8`

⚠️ The data will not be processed for compression by the library and the resource will be marked as having been compressed using ZLIB; accepted by SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) and SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) only, which also take the decompressed size (use with caution)

`ForceInternal = 4`

The data will be compressed using Maxis' internal method

`ForceOff = 1`

The data will not be compressed

`[Obsolete("This setting can no longer be used and will be removed in a future version. If you need to force a compression, use a different force member of this enum.", true)] ForceOn = 2`

Deprecated: do not use

`ForceZLib = 3`

The data will be compressed using ZLib

`SetDeletedFlag = 5`

⚠️ The data will not be processed for compression by the library—thus callers will need to invoke ZLibCompress(ReadOnlyMemory<byte>) or ZLibCompressAsync(ReadOnlyMemory<byte>, CancellationToken) themselves and pass the result to Set(ResourceKey, ReadOnlyMemory<byte>, CompressionMode) or SetAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, CancellationToken), respectively, if they intend for the content to be compressed with ZLib despite being flagged—and the deleted flag will be set (use with caution)

Remarks

Important: While this library does support the older compression methods, it does not have a complete catalog of how to do so properly for every resource type automatically (Auto). If you want to be sure that you're compressing a resource correctly, utilize GetExplicitCompressionMode(ResourceKey) with a package the game is currently accepting.

Warning: The Sims 4 does not always expect or tolerate ZLib compression for certain types of resources, such as for world package files which ship with the game (see thepancake1's comment about this on the Creator Musings Discord for more information).

Class DataBasePackedFile

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents a Maxis DBPF format package file (🔒)

```
public sealed class DataBasePackedFile : IAsyncDisposable, IDisposable
```

Inheritance

object ← DataBasePackedFile

Implements

IAsyncDisposable, IDisposable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Streams and Disposal

When a package is opened by this class from a Stream, the class assumes ownership of the Stream. Manipulating the Stream afterward may result in undefined behavior. The class will dispose of the Stream when it is, itself, disposed.

To use the Save(bool, ResourceKeyOrder) or SaveAsync(bool, ResourceKeyOrder, CancellationToken) methods, the package must have been opened from a Stream which is writeable. This can also be determined by the caller in advance of attempting to use them by checking the value of the CanSaveInPlace property. However, for ease of use in web frameworks like ASP.NET Core, the CopyTo(Stream, ResourceKeyOrder) and CopyToAsync(Stream, ResourceKeyOrder, CancellationToken) methods do *not* require that the Stream which they are passed is seekable.

Lazy Loading

When a package is first opened using this class, the package index is read but the content of the package resources is not. All of the retrieval methods of this class have async counterparts for callers who prefer not to be blocked by I/O operations which may be required when retrieving resource content.

Because resource names (when applicable) are determined by their content, this class will not index resource names until they are first referenced, either explicitly by calling GetNames() or GetNamesAsync(CancellationToken), or implicitly by calling any of the other methods which deal with resource names. Afterward, the names are cached for the lifetime of the package instance and updated as the caller sets the content of resources or deletes them.

Thread Safety

All properties and methods of this class, including async methods, are kept thread-safe with internally managed locks. Async methods which accept cancellation tokens will honor signals so long as they have not progressed to mutating the state of the package.

Compression

Decompression of resource content is handled automatically by this class.

When retrieving resource content, this class will automatically decompress it using either proprietary Maxis internal or streaming compression, or ZLib compression. When saving resources, this class will always use ZLib compression. For more details, see the remarks for the CompressionMode enumeration.

Resource content is compressed in memory when it is set to minimize the amount of memory used while operating on a package.

Deleted Entries

This class honors the deleted entry flag in the package index under normal circumstances. If a caller attempts to retrieve the content of a resource flagged as deleted, the class will throw a FileNotFoundException by default. All retrieval methods have an optional `force` bool parameter which, when true, will cause the class to attempt to retrieve the content of a resource flagged as deleted (this is not guaranteed to work).

Constructors

DataBasePackedFile()

Initializes a new, blank DataBasePackedFile

```
public DataBasePackedFile()
```

DataBasePackedFile(Stream)

Initializes a DataBasePackedFile from the specified `stream` (🗑️ 💤)

```
public DataBasePackedFile(Stream stream)
```

Parameters

`stream` Stream

DataBasePackedFile(string, bool)

Initializes a DataBasePackedFile from the specified `path` (🗑️ 💤)

```
public DataBasePackedFile(string path, bool forReadOnly = true)
```

Parameters

`path` string

`forReadOnly` bool

Properties

CanSaveInPlace

Gets whether the package can be saved in place

```
public bool CanSaveInPlace { get; }
```

Property Value

bool

Count

Gets the number of resources

```
public int Count { get; }
```

Property Value

int

CreationTime

Gets/sets when the package was created

```
public DateTimeOffset CreationTime { get; set; }
```

Property Value

DateTimeOffset

FileVersion

Gets/sets the version of the format of the file

```
public Version FileVersion { get; set; }
```

Property Value

Version

Remarks

This class is intended to work with versions of the Maxis DBPF format up to version `2.1` and going back as far as possible. While it will permit the caller to specify any version they like, it does not guard against potential negative consequences which may result from such packages actually being used. The default version for newly created packages is currently `2.1`. Versions `1.x` (used by The Sims 2) are written with the `1.x` header and index layouts, compress only with Maxis' internal algorithm, and record which resources are compressed in a directory resource (type, group, and instance `E86B1EEF`, `E86B1EEF`, `286B1F03`) which this class brings up to date whenever it writes the package, removing it when no resources are compressed.

Exceptions

InvalidOperationException

> The new value would move the package between version `1.x` and any other version while it contains a compressed or deleted resource (set those resources with ForceOff first)

this[ResourceKey]

Gets/sets the content of a resource with the specified `key`

```
public object? this[ResourceKey key] { get; set; }
```

Parameters

`key` ResourceKey

Property Value

object

Keys

Gets a list of keys for all the resources in the package

```
public IReadOnlyList<ResourceKey> Keys { get; }
```

Property Value

IReadOnlyList<ResourceKey>

Unused1

Gets/sets the four bytes in the package header starting at index 14h

```
public ReadOnlyMemory<byte> Unused1 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused2

Gets/sets the four bytes in the package header starting at index 20h

```
public ReadOnlyMemory<byte> Unused2 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused3

Gets/sets the twelve bytes in the package header starting at index 30h

```
public ReadOnlyMemory<byte> Unused3 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused4

Gets/sets the four bytes in the package header starting at index 3Ch

```
public ReadOnlyMemory<byte> Unused4 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused5

Gets/sets the twenty-four bytes in the package header starting at index 48h

```
public ReadOnlyMemory<byte> Unused5 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

UpdatedTime

Gets/sets when the package was last updated

```
public DateTimeOffset UpdatedTime { get; set; }
```

Property Value

DateTimeOffset

UserVersion

Gets/sets the version of the user (presumably the user agent, actually? -- is Maxis versioning human beings?)

```
public Version UserVersion { get; set; }
```

Property Value

Version

Methods

ContainsKey(ResourceKey)

Gets whether the package contains a resource with the specified `key`

```
public bool ContainsKey(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

bool

ContainsKeyAsync(ResourceKey, CancellationToken)

Gets whether the package contains a resource with the specified `key` asynchronously

```
public Task<bool> ContainsKeyAsync(ResourceKey key, CancellationToken cancellationToken
= default)
```

Parameters

`key` ResourceKey

`cancellationToken` CancellationToken

Returns

Task<bool>

CopyTo(Stream, ResourceKeyOrder)

Copies the package in binary format to the specified `destination` (🔁 🏃)

```
public void CopyTo(Stream destination, ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`destination` Stream

`resourceKeyOrder` ResourceKeyOrder

Exceptions

ArgumentNullException

`destination` is null

ArgumentException

`destination` is not writeable

CopyToAsync(Stream, ResourceKeyOrder, CancellationToken)

Copies the package in binary format to the specified `destination` asynchronously (🔄 🏃)

```
public Task CopyToAsync(Stream destination, ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`destination` Stream

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task

Exceptions

ArgumentNullException

`destination` is null

ArgumentException

`destination` is not writeable

Delete(ResourceKey)

Deletes the resource with the specified `key` and returns true if it was found; otherwise, false

```
public bool Delete(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

bool

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

DisposeAsync()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources asynchronously.

```
public ValueTask DisposeAsync()
```

Returns

ValueTask

A task that represents the asynchronous dispose operation.

~DataBasePackedFile()

Called by the finalizer

```
protected ~DataBasePackedFile()
```

ForEach(ResourceKeyOrder, Predicate<ResourceKey>, Action<ResourceKey, ReadOnlyMemory<byte>>, CancellationToken)

Processes undeleted resources in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAction`

```
public void ForEach(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Action<ResourceKey, ReadOnlyMemory<byte>> processResourceAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAction` Action<ResourceKey, ReadOnlyMemory<byte>>

`cancellationToken` CancellationToken

ForEachAsync(ResourceKeyOrder, Predicate<ResourceKey>, Func<ResourceKey, ReadOnlyMemory<byte>, Task>, CancellationToken)

Processes undeleted resources in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAsyncAction`

```
public Task ForEachAsync(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Func<ResourceKey, ReadOnlyMemory<byte>, Task> processResourceAsyncAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAsyncAction` Func<ResourceKey, ReadOnlyMemory<byte>, Task>

`cancellationToken` CancellationToken

Returns

Task

ForEachRaw(ResourceKeyOrder, Predicate<ResourceKey>, Action<ResourceKey, ReadOnlyMemory<byte>>, CancellationToken)

Processes undeleted raw resources (meaning that the library will not decompress a resource for you if it is compressed) in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAction`

```
public void ForEachRaw(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Action<ResourceKey, ReadOnlyMemory<byte>> processResourceAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAction` Action<ResourceKey, ReadOnlyMemory<byte>>

`cancellationToken` CancellationToken

ForEachRawAsync(ResourceKeyOrder, Predicate<ResourceKey>, Func<ResourceKey, ReadOnlyMemory<byte>, Task>, CancellationToken)

Processes undeleted raw resources (meaning that the library will not decompress a resource for you if it is compressed) in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAsyncAction`

```
public Task ForEachRawAsync(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Func<ResourceKey, ReadOnlyMemory<byte>, Task> processResourceAsyncAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAsyncAction` Func<ResourceKey, ReadOnlyMemory<byte>, Task>

`cancellationToken` CancellationToken

Returns

Task

FromPathAsync(string, bool)

Initializes a DataBasePackedFile asynchronously from the specified `path` (🗑️ 💤)

```
public static Task<DataBasePackedFile> FromPathAsync(string path, bool forReadOnly = true)
```

Parameters

`path` string

`forReadOnly` bool

Returns

Task<DataBasePackedFile>

FromStreamAsync(Stream, CancellationToken)

Initializes a DataBasePackedFile asynchronously from the specified `stream` (🗑️ 💤)

```
public static Task<DataBasePackedFile> FromStreamAsync(Stream stream, CancellationToken
cancellationToken = default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<DataBasePackedFile>

Get(ResourceKey, bool)

Gets the content of a resource with the specified `key`

```
public ReadOnlyMemory<byte> Get(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

ReadOnlyMemory<byte>

GetAllSizes(Predicate<ResourceKey>?)

Gets the size of the content of each resource in the package indexed by key

```
public IReadOnlyDictionary<ResourceKey, int> GetAllSizes(Predicate<ResourceKey>? predicate
= null)
```

Parameters

`predicate` Predicate<ResourceKey>

An optional predicate to filter the keys

Returns

IReadOnlyDictionary<ResourceKey, int>

GetAllSizesAsync(Predicate<ResourceKey>?, CancellationToken)

Gets the size of the content of each resource in the package indexed by key, asynchronously

```
public Task<IReadOnlyDictionary<ResourceKey, int>> GetAllSizesAsync(Predicate<ResourceKey>?
predicate = null, CancellationToken cancellationToken = default)
```

Parameters

`predicate` Predicate<ResourceKey>

An optional predicate to filter the keys

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<IReadOnlyDictionary<ResourceKey, int>>

GetAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` asynchronously

```
public Task<ReadOnlyMemory<byte>> GetAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

GetCombinedTuning(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a CombinedTuningModel

```
public CombinedTuningModel GetCombinedTuning(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

CombinedTuningModel

GetCombinedTuningAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a CombinedTuningModel asynchronously

```
public Task<CombinedTuningModel> GetCombinedTuningAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<CombinedTuningModel>

GetCount()

Gets the number of resources

```
public int GetCount()
```

Returns

int

GetCountAsync(CancellationToken)

Gets the number of resources asychronously

```
public Task<int> GetCountAsync(CancellationToken cancellationToken = default)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task<int>

GetDdsAsPng(ResourceKey, bool)

Gets the content of a resource that is a DirectDraw Surface with the specified `key`, encoded as a PNG

```
public ReadOnlyMemory<byte> GetDdsAsPng(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

ReadOnlyMemory<byte>

GetDdsAsPngAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource that is a DirectDraw Surface with the specified `key`, encoded as a PNG, asynchronously

```
public Task<ReadOnlyMemory<byte>> GetDdsAsPngAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

GetExplicitCompressionMode(ResourceKey)

Gets the CompressionMode value to pass to Set methods to ensure they will compress the content of the resource specified by `key` in the same way it is currently compressed in the package

```
public CompressionMode GetExplicitCompressionMode(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

CompressionMode

Remarks

For a resource compressed using Maxis' streamable method, which this library cannot produce, this returns CallerSuppliedStreamable, which must be passed with the resource's raw content to SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) rather than to Set methods.

GetKeys(ResourceKeyOrder)

Gets a list of keys for all the resources in the package

```
public IReadOnlyList<ResourceKey> GetKeys(ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`resourceKeyOrder` ResourceKeyOrder

Returns

IReadOnlyList<ResourceKey>

GetKeysAsync(ResourceKeyOrder, CancellationToken)

Gets a list of keys for all the resources in the package asynchronously

```
public Task<IReadOnlyList<ResourceKey>> GetKeysAsync(ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task<IReadOnlyList<ResourceKey>>

GetKeysByName(string)

Gets a list of keys for resources with the specified `name`

```
public IReadOnlyList<ResourceKey> GetKeysByName(string name)
```

Parameters

`name` string

Returns

IReadOnlyList<ResourceKey>

GetKeysByNameAsync(string, CancellationToken)

Gets a list of keys for resources with the specified `name` asynchronously

```
public Task<IReadOnlyList<ResourceKey>> GetKeysByNameAsync(string name, CancellationToken
cancellationToken = default)
```

Parameters

`name` string

`cancellationToken` CancellationToken

Returns

Task<IReadOnlyList<ResourceKey>>

GetModFileManifest(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a ModFileManifestModel

```
public ModFileManifestModel GetModFileManifest(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

ModFileManifestModel

GetModFileManifestAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a ModFileManifestModel asynchronously

```
public Task<ModFileManifestModel> GetModFileManifestAsync(ResourceKey key, bool force =
false, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<ModFileManifestModel>

GetModel(ResourceKey, bool)

Gets the content of a resource with the specified `key` as the model GetModelType(ResourceType) gives for its type

```
public Model GetModel(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

Model

Exceptions

ArgumentException

No model decodes resources of the type specified by `key`

GetModelAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as the model GetModelType(ResourceType) gives for its type asynchronously

```
public Task<Model> GetModelAsync(ResourceKey key, bool force = false, CancellationToken
cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<Model>

Exceptions

ArgumentException

No model decodes resources of the type specified by `key`

GetModelAsync<TModel>(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a `TModel` asynchronously

```
public Task<TModel> GetModelAsync<TModel>(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default) where TModel : IModel<TModel>
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<TModel>

Type Parameters

`TModel`

A model used to browse and modify certain types of resources

Exceptions

ArgumentException

`TModel` cannot deal with resources of the type specified by `key`

GetModel<TModel>(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a `TModel`

```
public TModel GetModel<TModel>(ResourceKey key, bool force = false) where TModel
: IModel<TModel>
```

Parameters

`key` ResourceKey

`force` bool

Returns

TModel

Type Parameters

`TModel`

A model used to browse and modify certain types of resources

Exceptions

ArgumentException

`TModel` cannot deal with resources of the type specified by `key`

GetNameByKey(ResourceKey)

Gets the name for the resource with the specified `key`

```
public string? GetNameByKey(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

string

GetNameByKeyAsync(ResourceKey, CancellationToken)

Gets the name for the resource with the specified `key` asynchronously

```
public Task<string?> GetNameByKeyAsync(ResourceKey key, CancellationToken cancellationToken
= default)
```

Parameters

`key` ResourceKey

`cancellationToken` CancellationToken

Returns

Task <string>

GetNames()

Gets the names for all the resources in the package

```
public IReadOnlyList<string> GetNames()
```

Returns

IReadOnlyList <string>

GetNamesAsync(CancellationToken)

Gets the names for all the resources in the package asynchronously

```
public Task<IReadOnlyList<string>> GetNamesAsync(CancellationToken cancellationToken
= default)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task<IReadOnlyList<string>>

GetRaw(ResourceKey, bool)

Gets the raw content of a resource with the specified `key`, meaning that the library will not decompress it for you if it is compressed

```
public ReadOnlyMemory<byte> GetRaw(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

ReadOnlyMemory<byte>

GetRawAsync(ResourceKey, bool, CancellationToken)

Gets the raw content of a resource with the specified `key` asynchronously, meaning that the library will not decompress it for you if it is compressed

```
public Task<ReadOnlyMemory<byte>> GetRawAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

GetSimData(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a SimDataModel

```
public SimDataModel GetSimData(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

SimDataModel

GetSimDataAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a SimDataModel asynchronously

```
public Task<SimDataModel> GetSimDataAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task <SimDataModel>

GetSize(ResourceKey)

Gets the size of the content of the resource with the specified `key`

```
public int GetSize(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

int

GetSizeAsync(ResourceKey, CancellationToken)

Gets the size of the content of the resource with the specified `key` asynchronously

```
public Task<int> GetSizeAsync(ResourceKey key, CancellationToken cancellationToken
```

```
= default)
```

Parameters

`key` ResourceKey

`cancellationToken` CancellationToken

Returns

Task<int>

GetStringTable(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a StringTableModel

```
public StringTableModel GetStringTable(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

StringTableModel

GetStringTableAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a StringTableModel asynchronously

```
public Task<StringTableModel> GetStringTableAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<StringTableModel>

GetText(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a string

```
public string GetText(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

string

GetTextAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a string asynchronously

```
public Task<string> GetTextAsync(ResourceKey key, bool force = false, CancellationToken
cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<string>

GetTranslucentJpegAsPng(ResourceKey, bool)

Gets the content of a resource that is a TS4 proprietary translucent JPEG based with the specified `key`, reintegrated into RGBA and encoded as a PNG

```
public ReadOnlyMemory<byte> GetTranslucentJpegAsPng(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

ReadOnlyMemory<byte>

GetTranslucentJpegAsPngAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource that is a TS4 proprietary translucent JPEG based with the specified `key`, reintegrated into RGBA and encoded as a PNG, asynchronously

```
public Task<ReadOnlyMemory<byte>> GetTranslucentJpegAsPngAsync(ResourceKey key, bool force =
```

```
false, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

GetXml(ResourceKey, bool)

Gets the content of a resource with the specified `key` as an XDocument

```
public XDocument GetXml(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

XDocument

GetXmlAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as an XDocument asynchronously

```
public Task<XDocument> GetXmlAsync(ResourceKey key, bool force = false, CancellationToken
cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<XDocument>

InternalCompress(ReadOnlyMemory<byte>, out bool)

Compresses the specified `memory` in the manner of Maxis' internal algorithm, returning the result

```
[SuppressMessage("Design", "CA1021: Avoid out parameters")]
public static ReadOnlyMemory<byte> InternalCompress(ReadOnlyMemory<byte> memory, out
bool wasCompressed)
```

Parameters

`memory` ReadOnlyMemory<byte>

`wasCompressed` bool

Returns

ReadOnlyMemory<byte>

InternalDecompress(ReadOnlyMemory<byte>)

Decompresses the specified `memory` in the manner of Maxis' internal algorithm, returning the result

```
public static ReadOnlyMemory<byte> InternalDecompress(ReadOnlyMemory<byte> memory)
```

Parameters

`memory` ReadOnlyMemory<byte>

Returns

ReadOnlyMemory<byte>

LoadAll(bool, CompressionMode)

Loads all resources in the package into memory

```
public void LoadAll(bool force = false, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`force` bool

true to get the content of the resources even if they has been marked as deleted; otheriwse, false (default)

`compressionMode` CompressionMode

This parameter is now ignored

LoadAllAsync(bool, CompressionMode, CancellationToken)

Loads all resources in the package into memory

```
public Task LoadAllAsync(bool force = false, CompressionMode compressionMode =
CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`force` bool

true to get the content of the resources even if they has been marked as deleted; otheriwse, false (default)

`compressionMode` CompressionMode

This parameter is now ignored

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

LoadNames()

Loads the names of all the resources in the package (a no-op if the names are already loaded)

```
public void LoadNames()
```

LoadNamesAsync(CancellationToken)

Loads the names of all the resources in the package asynchronously (a no-op if the names are already loaded)

```
public Task LoadNamesAsync(CancellationToken cancellationToken = default)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task

Save(bool, ResourceKeyOrder)

Saves the package to the stream from which it was loaded

```
public void Save(bool unloadFromMemory = false, ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`unloadFromMemory` bool

`resourceKeyOrder` ResourceKeyOrder

Remarks

This method overwrites the stream in place. If it fails partway through writing (for example, because the disk is full, another process has the file open, or the process is terminated), the stream will hold neither the original package nor the new one, and the original cannot be recovered. If losing the original would be costly, load the package as read-only, use SaveAs(string, ResourceKeyOrder) to write a temporary file in the same directory, dispose of the package, and then replace the original with the temporary file (e.g. with Replace(string, string, string)). SaveAs(string, ResourceKeyOrder) cannot write to the path from which the package is currently open.

Exceptions

InvalidOperationException

> The package was not loaded from stream or the stream is not writeable

SaveAs(string, ResourceKeyOrder)

Saves the package to the specified `path` (🔄 🏃)

```
public void SaveAs(string path, ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`path` string

`resourceKeyOrder` ResourceKeyOrder

SaveAsAsync(string, ResourceKeyOrder, CancellationToken)

Saves the package to the specified `path` asynchronously (🔄 🏃)

```
public Task SaveAsAsync(string path, ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`path` string

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task

SaveAsync(bool, ResourceKeyOrder, CancellationToken)

Saves the package to the stream from which it was loaded asynchronously

```
public Task SaveAsync(bool unloadFromMemory = false, ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`unloadFromMemory` bool

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task

Remarks

This method overwrites the stream in place. If it fails partway through writing (for example, because the disk is full, another process has the file open, or the process is terminated), the stream will hold neither the original package nor the new one, and the original cannot be recovered. If losing the original would be costly, load the package as read-only, use SaveAsAsync(string, ResourceKeyOrder, CancellationToken) to write a temporary file in the same directory, dispose of the package, and then replace the original with the temporary file (e.g. with Replace(string, string, string)). SaveAsAsync(string, ResourceKeyOrder, CancellationToken) cannot write to the path from which the package is currently open.

Exceptions

InvalidOperationException

The package was not loaded from stream or the stream is not writeable

Set(ResourceKey, IModel, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, IModel content, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` IModel

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

Set(ResourceKey, ReadOnlyMemory<byte>, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, ReadOnlyMemory<byte> content, CompressionMode
compressionMode = CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` ReadOnlyMemory<byte>

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

Set(ResourceKey, string, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, string content, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` string

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

Set(ResourceKey, XDocument, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, XDocument content, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` XDocument

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

SetAsync(ResourceKey, IModel, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, IModel content, CompressionMode compressionMode
= CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` IModel

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

SetAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, ReadOnlyMemory<byte> content, CompressionMode
compressionMode = CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` ReadOnlyMemory<byte>

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task <bool>

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

SetAsync(ResourceKey, string, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, string content, CompressionMode compressionMode
= CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` string

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task <bool>

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

SetAsync(ResourceKey, XDocument, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, XDocument content, CompressionMode
compressionMode = CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` XDocument

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

Exceptions

ArgumentException

> `compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

SetPngAsDds(ResourceKey, ReadOnlyMemory<byte>)

Sets the content of a resource specified by `key` to a shuffled DirectDraw Surface based on `pngData`

```
public void SetPngAsDds(ResourceKey key, ReadOnlyMemory<byte> pngData)
```

Parameters

`key` ResourceKey

> The key of the resource

`pngData` ReadOnlyMemory<byte>

> The encoded PNG data

SetPngAsDdsAsync(ResourceKey, ReadOnlyMemory<byte>, CancellationToken)

Sets the content of a resource specified by `key` to a shuffled DirectDraw Surface based on `pngData` asycnhronously

```
public Task SetPngAsDdsAsync(ResourceKey key, ReadOnlyMemory<byte> pngData,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

SetPngAsTranslucentJpeg(ResourceKey, ReadOnlyMemory<byte>)

Sets the content of a resource specified by `key` to a TS4 proprietary translucent JPEG based on `pngData`

```
public void SetPngAsTranslucentJpeg(ResourceKey key, ReadOnlyMemory<byte> pngData)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

SetPngAsTranslucentJpegAsync(ResourceKey, ReadOnlyMemory<byte>, CancellationToken)

Sets the content of a resource specified by `key` to a TS4 proprietary translucent JPEG based on `pngData` asycnhronously

```
public Task SetPngAsTranslucentJpegAsync(ResourceKey key, ReadOnlyMemory<byte> pngData,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int)

Sets the raw content of a resource with the specified `key`, meaning that the library stores `rawContent` as it is, marked as compressed in the manner of `compressionMode`, with the decompressed size the caller states

```
public void SetRaw(ResourceKey key, ReadOnlyMemory<byte> rawContent, CompressionMode
compressionMode, int sizeDecompressed)
```

Parameters

`key` ResourceKey

The key of the resource

`rawContent` ReadOnlyMemory<byte>

The content of the resource, already compressed by the caller (for DBPF 1.x packages, including the four-byte size prefix that format requires)

`compressionMode` CompressionMode

The manner in which `rawContent` is compressed: CallerSuppliedInternal, CallerSuppliedStreamable, or CallerSuppliedZLib

`sizeDecompressed` int

The size of the content of the resource once decompressed

Exceptions

ArgumentException

`compressionMode` is not a caller-supplied compression mode

ArgumentOutOfRangeException

`sizeDecompressed` is negative

NotSupportedException

The package is DBPF 1.x and `compressionMode` is not CallerSuppliedInternal

SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken)

Sets the raw content of a resource with the specified `key` asynchronously, meaning that the library stores `rawContent` as it is, marked as compressed in the manner of `compressionMode`, with the decompressed size the caller states

```
public Task SetRawAsync(ResourceKey key, ReadOnlyMemory<byte> rawContent, CompressionMode
compressionMode, int sizeDecompressed, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`rawContent` ReadOnlyMemory<byte>

The content of the resource, already compressed by the caller (for DBPF 1.x packages, including the four-byte size prefix that format requires)

`compressionMode` CompressionMode

The manner in which `rawContent` is compressed: CallerSuppliedInternal, CallerSuppliedStreamable, or CallerSuppliedZLib

`sizeDecompressed` int

The size of the content of the resource once decompressed

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

Exceptions

ArgumentException

`compressionMode` is not a caller-supplied compression mode

ArgumentOutOfRangeException

`sizeDecompressed` is negative

NotSupportedException

The package is DBPF 1.x and `compressionMode` is not CallerSuppliedInternal

SetXml(ResourceKey, string, CompressionMode)

Sets the `xmlContent` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool SetXml(ResourceKey key, string xmlContent, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`xmlContent` string

The XML content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

SetXmlAsync(ResourceKey, string, CompressionMode, CancellationToken)

Sets the `xmlContent` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetXmlAsync(ResourceKey key, string xmlContent, CompressionMode
compressionMode = CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`xmlContent` string

The XML content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

Exceptions

ArgumentException

`compressionMode` is a caller-supplied compression mode (use SetRaw(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int) or SetRawAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, int, CancellationToken) instead)

ZLibCompress(ReadOnlyMemory<byte>)

Compresses the specified `memory` using ZLib, returning the result

```
public static ReadOnlyMemory<byte> ZLibCompress(ReadOnlyMemory<byte> memory)
```

Parameters

`memory` ReadOnlyMemory<byte>

Returns

ReadOnlyMemory<byte>

ZLibCompressAsync(ReadOnlyMemory<byte>, CancellationToken)

Compresses the specified `memory` using ZLib asynchronously, returning the result

```
public static Task<ReadOnlyMemory<byte>> ZLibCompressAsync(ReadOnlyMemory<byte> memory,
CancellationToken cancellationToken = default)
```

Parameters

`memory` ReadOnlyMemory<byte>

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

ZLibDecompress(ReadOnlyMemory<byte>, int)

Decompresses the specified `memory` using ZLib, returning the result at the specified `sizeDecompressed`

```
public static ReadOnlyMemory<byte> ZLibDecompress(ReadOnlyMemory<byte> memory,
int sizeDecompressed)
```

Parameters

`memory` ReadOnlyMemory<byte>

`sizeDecompressed` int

Returns

ReadOnlyMemory<byte>

ZLibDecompressAsync(ReadOnlyMemory<byte>, int, CancellationToken)

Decompresses the specified `memory` using ZLib asynchronously, returning the result at the specified `sizeDecompressed`

```
public static Task<ReadOnlyMemory<byte>> ZLibDecompressAsync(ReadOnlyMemory<byte> memory,
```

```
  int sizeDecompressed, CancellationToken cancellationToken = default)
```

Parameters

`memory` ReadOnlyMemory<byte>

`sizeDecompressed` int

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

Class ReadOnlyMemoryOfByteStream

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents a read-only, seekable Stream over a ReadOnlyMemory<T> of bytes (🔓)

```
public sealed class ReadOnlyMemoryOfByteStream : Stream, IAsyncDisposable, IDisposable
```

Inheritance

object ← MarshalByRefObject ← Stream ← ReadOnlyMemoryOfByteStream

Implements

IAsyncDisposable, IDisposable

Inherited Members

Stream.Null , Stream.BeginRead(byte[], int, int, AsyncCallback, object) , Stream.BeginWrite(byte[], int, int, AsyncCallback, object) , Stream.Close() , Stream.CopyTo(Stream) , Stream.CopyTo(Stream, int) , Stream.CopyToAsync(Stream) , Stream.CopyToAsync(Stream, int) , Stream.CopyToAsync(Stream, int, CancellationToken) , Stream.CopyToAsync(Stream, CancellationToken) , Stream.Dispose() , Stream.DisposeAsync() , Stream.EndRead(IAsyncResult) , Stream.EndWrite(IAsyncResult) , Stream.FlushAsync() , Stream.FlushAsync(CancellationToken) , Stream.Read(Span<byte>) , Stream.ReadAsync(byte[], int, int) , Stream.ReadAsync(byte[], int, int, CancellationToken) , Stream.ReadAsync(Memory<byte>, CancellationToken) , Stream.ReadByte() , Stream.Synchronized(Stream) , Stream.Write(ReadOnlySpan<byte>) , Stream.WriteAsync(byte[], int, int) , Stream.WriteAsync(byte[], int, int, CancellationToken) , Stream.WriteAsync(ReadOnlyMemory<byte>, CancellationToken) , Stream.WriteByte(byte) , Stream.CanTimeout , Stream.ReadTimeout , Stream.WriteTimeout , MarshalByRefObject.GetLifetimeService() , MarshalByRefObject.InitializeLifetimeService() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ReadOnlyMemoryOfByteStream(ReadOnlyMemory<byte>)

Initializes a new ReadOnlyMemoryOfByteStream over the specified `readOnlyMemoryOfBytes`

```
public ReadOnlyMemoryOfByteStream(ReadOnlyMemory<byte> readOnlyMemoryOfBytes)
```

Parameters

`readOnlyMemoryOfBytes` ReadOnlyMemory<byte>

Properties

CanRead

When overridden in a derived class, gets a value indicating whether the current stream supports reading.

```
public override bool CanRead { get; }
```

Property Value

bool

true if the stream supports reading; otherwise, false.

CanSeek

When overridden in a derived class, gets a value indicating whether the current stream supports seeking.

```
public override bool CanSeek { get; }
```

Property Value

bool

true if the stream supports seeking; otherwise, false.

CanWrite

When overridden in a derived class, gets a value indicating whether the current stream supports writing.

```
public override bool CanWrite { get; }
```

Property Value

bool

true if the stream supports writing; otherwise, false.

Length

When overridden in a derived class, gets the length in bytes of the stream.

```
public override long Length { get; }
```

Property Value

long

A long value representing the length of the stream in bytes.

Exceptions

NotSupportedException

A class derived from `Stream` does not support seeking.

ObjectDisposedException

Methods were called after the stream was closed.

Memory

Gets the bytes the stream reads

```
public ReadOnlyMemory<byte> Memory { get; }
```

Property Value

ReadOnlyMemory<byte>

Position

When overridden in a derived class, gets or sets the position within the current stream.

```
public override long Position { get; set; }
```

Property Value

long

The current position within the stream.

Exceptions

IOException

An I/O error occurs.

NotSupportedException

The stream does not support seeking.

ObjectDisposedException

Methods were called after the stream was closed.

Methods

Flush()

When overridden in a derived class, clears all buffers for this stream and causes any buffered data to be written to the underlying device.

```
public override void Flush()
```

Exceptions

IOException

An I/O error occurs.

Read(byte[], int, int)

When overridden in a derived class, reads a sequence of bytes from the current stream and advances the position within the stream by the number of bytes read.

```
public override int Read(byte[] buffer, int offset, int count)
```

Parameters

`buffer` byte[]

An array of bytes. When this method returns, the buffer contains the specified byte array with the values between `offset` and (`offset` + `count` - 1) replaced by the bytes read from the current source.

`offset` int

The zero-based byte offset in `buffer` at which to begin storing the data read from the current stream.

`count` int

The maximum number of bytes to be read from the current stream.

Returns

int

The total number of bytes read into the buffer. This can be less than the number of bytes requested if that many bytes are not currently available, or zero (0) if the end of the stream has been reached.

Exceptions

ArgumentException

The sum of `offset` and `count` is larger than the buffer length.

ArgumentNullException

`buffer` is null.

ArgumentOutOfRangeException

`offset` or `count` is negative.

IOException

An I/O error occurs.

NotSupportedException

The stream does not support reading.

ObjectDisposedException

Methods were called after the stream was closed.

Seek(long, SeekOrigin)

When overridden in a derived class, sets the position within the current stream.

```
public override long Seek(long offset, SeekOrigin origin)
```

Parameters

`offset` long

A byte offset relative to the `origin` parameter.

`origin` SeekOrigin

A value of type SeekOrigin indicating the reference point used to obtain the new position.

Returns

long

The new position within the current stream.

Exceptions

IOException

An I/O error occurs.

NotSupportedException

The stream does not support seeking, such as if the stream is constructed from a pipe or console output.

ObjectDisposedException

Methods were called after the stream was closed.

SetLength(long)

When overridden in a derived class, sets the length of the current stream.

```
public override void SetLength(long value)
```

Parameters

`value` long

The desired length of the current stream in bytes.

Exceptions

IOException

An I/O error occurs.

NotSupportedException

The stream does not support both writing and seeking, such as if the stream is constructed from a pipe or console output.

ObjectDisposedException

Methods were called after the stream was closed.

Write(byte[], int, int)

When overridden in a derived class, writes a sequence of bytes to the current stream and advances the current position within this stream by the number of bytes written.

```
public override void Write(byte[] buffer, int offset, int count)
```

Parameters

`buffer` byte[]

An array of bytes. This method copies `count` bytes from `buffer` to the current stream.

`offset` int

The zero-based byte offset in `buffer` at which to begin copying bytes to the current stream.

`count` int

The number of bytes to be written to the current stream.

Exceptions

ArgumentException

The sum of `offset` and `count` is greater than the buffer length.

ArgumentNullException

`buffer` is null.

ArgumentOutOfRangeException

`offset` or `count` is negative.

IOException

An I/O error occurred, such as the specified file cannot be found.

NotSupportedException

The stream does not support writing.

ObjectDisposedException

Write(byte[], int, int) was called after the stream was closed.

Enum ResourceFileType

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

The file type of a package resource independent of storage within a DataBasePackedFile

```
public enum ResourceFileType
```

Fields

`[ResourceFileTypeFormat("bin", "application/octet-stream")] Binary = 0`

Binary

`[ResourceFileTypeFormat("dds", "image/vnd.ms-dds")] DirectDrawSurface = 1`

DDS Image

`[ResourceFileTypeFormat("png", "image/png")] DirectDrawSurfaceAsPortableNetworkGraphic = 2`

PNG Image converted from DDS Image

`[ResourceFileTypeFormat("json", "application/json")] JavaScriptObjectNotation = 3`

JavaScript Object Notation

`[ResourceFileTypeFormat("jpg", "image/jpeg")] JointPhotographicExpertsGroupImage = 4`

JPEG Image

`[ResourceFileTypeFormat("png", "image/png")] PortableNetworkGraphic = 5`

PNG Image

`[ResourceFileTypeFormat("jpg", "image/jpeg")] Ts4TranslucentJointPhotographicExpertsGroupImage = 7`

The Sims 4's proprietary image format in which a PNG alpha mask is embedded in the headers of a JPEG

```
[ResourceFileTypeFormat("xml", "application/xml")] TuningMarkup = 6
```

Tuning XML

```
[ResourceFileTypeFormat("yml", "application/x-yaml")] YetAnotherMarkupLanguage = 8
```

Yet Another Markup Language

Class ResourceFileTypeAttribute

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies the file type of a package resource type independent of storage within a DataBasePackedFile

```
[AttributeUsage(AttributeTargets.Field, AllowMultiple = false)]
public sealed class ResourceFileTypeAttribute : Attribute
```

Inheritance

object ← Attribute ← ResourceFileTypeAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) , Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) , Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) , Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) , Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) ,
object.ToString()

Constructors

ResourceFileTypeAttribute(ResourceFileType)

Specifies the file type of a package resource type independent of storage within a DataBasePackedFile

```
public ResourceFileTypeAttribute(ResourceFileType packageResourceFileType)
```

Parameters

`packageResourceFileType` ResourceFileType

Properties

ResourceFileType

Gets the file type of the decorated package resource type independent of storage within a DataBase PackedFile

```
public ResourceFileType ResourceFileType { get; }
```

Property Value

ResourceFileType

Class ResourceFileTypeFormatAttribute

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies the extension and mime type of a file type of a package resource independent of storage within a DataBasePackedFile

```
[AttributeUsage(AttributeTargets.Field, AllowMultiple = false)]
public sealed class ResourceFileTypeFormatAttribute : Attribute
```

Inheritance

object ← Attribute ← ResourceFileTypeFormatAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) , Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) , Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) , Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) , Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) ,
object.ToString()

Constructors

ResourceFileTypeFormatAttribute(string, string?)

Specifies the extension and mime type of a file type of a package resource independent of storage within a DataBasePackedFile

```
public ResourceFileTypeFormatAttribute(string extension, string? mimeType = null)
```

Parameters

`extension` string

`mimeType` string

Properties

Extension

Gets the extention used for files in the format

```
public string Extension { get; }
```

Property Value

string

MimeType

Gets the MIME type of the format

```
public string? MimeType { get; }
```

Property Value

string

Struct ResourceKey

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents the unique key of a resource within a DataBasePackedFile

```
public struct ResourceKey : IEquatable<ResourceKey>
```

Implements
IEquatable <ResourceKey>

Inherited Members
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Extension Methods
SmartSimUtilities.GetResourceKeyForLocale(ResourceKey, CultureInfo) ,
SmartSimUtilities.GetStringTableLocale(ResourceKey)

Constructors

ResourceKey(ResourceType, uint, ulong)

Represents the unique key of a resource within a DataBasePackedFile

```
public ResourceKey(ResourceType type, uint group, ulong fullInstance)
```

Parameters

`type` ResourceType

`group` uint

`fullInstance` ulong

Fields

Empty

Represents the empty ResourceKey

```
public static readonly ResourceKey Empty
```

Field Value

ResourceKey

FullInstance

A value that uniquely identifies an individual resource within a given type and group

```
public readonly ulong FullInstance
```

Field Value

ulong

Group

A value used to group related resources together

```
public readonly uint Group
```

Field Value

uint

Remarks

In many cases, this value can be 0, especially for custom content. However, Maxis content and some specific modding scenarios might use different group values to organize resources. Group values help in categorizing resources that belong to a specific set or context.

Type

The type of resource indicated by the key

```
public readonly ResourceType Type
```

Field Value

ResourceType

Properties

FullInstanceHex

Gets the hex representation of FullInstance

```
public readonly string FullInstanceHex { get; }
```

Property Value

string

FullTgi

Gets the full TGI or TGI ID (type, group, and instance in hex format) of the resource key

```
public readonly string FullTgi { get; }
```

Property Value

string

GroupHex

Gets the hex representation of Group

```
public readonly string GroupHex { get; }
```

Property Value

string

Instance

Gets the low order bits of FullInstance

```
public readonly uint Instance { get; }
```

Property Value

uint

InstanceEx

Gets the high order bits of FullInstance

```
public readonly uint InstanceEx { get; }
```

Property Value

uint

InstanceExHex

Gets the hex representation of InstanceEx

```
public readonly string InstanceExHex { get; }
```

Property Value

string

InstanceHex

Gets the hex representation of Instance

```
public readonly string InstanceHex { get; }
```

Property Value

string

TypeHex

Gets the hex representation of Type

```
public readonly string TypeHex { get; }
```

Property Value

string

Methods

Deconstruct(out ResourceType, out uint, out ulong)

Deconstructs the key into its components

```
public readonly void Deconstruct(out ResourceType type, out uint group, out
ulong fullInstance)
```

Parameters

`type` ResourceType

The Type

`group` uint

The Group

`fullInstance` ulong

The FullInstance

Equals(ResourceKey)

Indicates whether the current object is equal to another object of the same type.

```
public readonly bool Equals(ResourceKey other)
```

Parameters

`other` ResourceKey

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Indicates whether this instance and a specified object are equal.

```
public override readonly bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override readonly int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

Parse(string)

Parses a string into a ResourceKey

```
public static ResourceKey Parse(string s)
```

Parameters

`s` string

The string to parse

Returns

ResourceKey

Exceptions

FormatException

`s` is not in the correct format

Parse(string, IFormatProvider?)

Parses a string into a ResourceKey

```
public static ResourceKey Parse(string s, IFormatProvider? provider)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

Returns

ResourceKey

Exceptions

FormatException

`s` is not in the correct format

ToString()

Returns the fully qualified type name of this instance.

```
public override readonly string ToString()
```

Returns

string

The fully qualified type name.

TryParse(string?, out ResourceKey)

Tries to parse a string into a ResourceKey

```
public static bool TryParse(string? s, out ResourceKey result)
```

Parameters

`s` string

The string to parse

`result` ResourceKey

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

TryParse(string?, IFormatProvider?, out ResourceKey)

Tries to parse a string into a ResourceKey

```
public static bool TryParse(string? s, IFormatProvider? provider, out ResourceKey result)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

`result` ResourceKey

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

Operators

operator ==(ResourceKey, ResourceKey)

```
public static bool operator ==(ResourceKey left, ResourceKey right)
```

Parameters

`left` ResourceKey

`right` ResourceKey

Returns

bool

implicit operator ResourceKey(string)

Converts a string into a ResourceKey

```
[SuppressMessage("Usage", "CA2225:Operator overloads have named alternates", Justification =
"For convenience in Python.")]
public static implicit operator ResourceKey(string s)
```

Parameters

`s` string

The string to convert

Returns

ResourceKey

operator !=(ResourceKey, ResourceKey)

```
public static bool operator !=(ResourceKey left, ResourceKey right)
```

Parameters

`left` ResourceKey

`right` ResourceKey

Returns

bool

Enum ResourceKeyOrder

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies the order in which a ResourceKey sequence should be processed

```
public enum ResourceKeyOrder
```

Fields

`InstanceTypeGroup = 1`

The operation will process the ResourceKey sequence in the order FullInstance, then Type, then Group

`Preserve = 0`

The operation will process the ResourceKey sequence in the order in which it was originally encountered

`TypeGroupInstance = 2`

The operation will process the ResourceKey sequence in the order Type, then Group, then FullInstance

Class ResourceToolingMetadataAttribute

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies that the ResourceType is tooling metadata and not intended for use by the game

```
[AttributeUsage(AttributeTargets.Field, AllowMultiple = false)]
public sealed class ResourceToolingMetadataAttribute : Attribute
```

Inheritance

object ← Attribute ← ResourceToolingMetadataAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) , Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) , Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) , Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) , Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) ,
object.ToString()

Enum ResourceType

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

A type of resource within a DataBasePackedFile

```
[SuppressMessage("Design", "CA1028: Enum Storage should be Int32", Justification = "This is by nature of how the package files work. We have no control over that.")]
public enum ResourceType : uint
```

Fields

`AVI = 929579223`

AVI

`[ResourceFileType(ResourceFileType.TuningMarkup)] AccountRewardTuning = 3138329048`

Account Reward Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] AchievementCategoryTuning = 609337601`

Achievement Category Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] AchievementCollectionTuning = 80917605`

Achievement Collection Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] AchievementTuning = 2018877086`

Achievement Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] ActionTuning = 209137191`

Action Tuning

`AgeGenderMap = 17805169`

Age Gender Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Ambience = 3537656749
```

Ambience

```
AnimationBoundaryConditionCache = 479834948
```

Animation Boundary Condition Cache

```
AnimationConstraintCache = 3794048034
```

Animation Constraint Cache

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AnimationStateMachine = 47570707
```

Animation State Machine

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AnimationTuning = 3994535597
```

Animation Tuning

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] ApartmentThumbnail1 = 2870590650
```

Apartment Thumbnail 1

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] ApartmentThumbnail2 = 3175683878
```

Apartment Thumbnail 2

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AspirationCategoryTuning =
3813727192
```

Aspiration Category Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AspirationTrackTuning =
3223387309
```

Aspiration Track Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AspirationTuning = 683034229
```

Aspiration Tuning

```
[ResourceFileType(ResourceFileType.Binary)] AudioConfiguration = 4244956094
```

Audio Configuration

```
AudioConfigurationReference = 968010314
```

Audio Configuration Reference

```
AudioEffects = 32437818
```

Audio Effects

```
AudioVocals = 27600859
```

Audio Vocals

```
AutomationTuning = 1421690392
```

Automation Tuning (the game's `Automation`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AwayActionTuning = 2947394632
```

Away Action Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BalloonTuning = 3966406598
```

Balloon Tuning

```
BatchFixHistory = 1810979774
```

Batch Fix History

```
[ResourceFileType(ResourceFileType.Binary)] BlendGeometry = 108833297
```

Blend Geometry

```
Block = 127102176
```

Block

```
Blueprint = 958717478
```

Blueprint

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] BlueprintImage = 3543935006
```

Blueprint Image

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] BodyPartThumbnail = 1529359685
```

Body Part Thumbnail

```
BonePose = 55959718
```

Bone Pose

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BreedTuning = 874331941
```

Breed Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BroadcasterTuning = 3736796019
```

Broadcaster Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BucksPerkTuning = 3963461902
```

Bucks Perk Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BuffTuning = 1612179606
```

Buff Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BuildBuyTuning = 3772208761
```

Build Buy Tuning (the game's `BuildBuy`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BusinessRuleTuning = 3102051436
```

Business Rule Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BusinessTuning = 1977092083
```

Business Tuning

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] BuyBuildThumbnail = 1009419847
```

Buy Build Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASCameraTuning = 626712290
```

CAS Camera Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASLightingTuning = 2148152976
```

CAS Lighting Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASMenuItemTuning = 213537012`

CAS Menu Item Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASMenuTuning = 2472182722`

CAS Menu Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASModifierTuning = 4088135484`

CAS Modifier Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASOccultSkintoneTuning = 2076404814`

CAS Occult Skintone Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASOccultTuning = 563571177`

CAS Occult Tuning

`[ResourceFileType(ResourceFileType.Binary)] CASPart = 55242443`

CAS Part

`[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)] CASPartThumbnail = 1008398834`

CAS Part Thumbnail

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASPreferenceCategoryTuning = 3456433227`

CAS Preference Category Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASPreferenceGroupTuning = 1720776484`

CAS Preference Group Tuning (the game's `CAS_PREFERENCE_GROUP`)

`[ResourceFileType(ResourceFileType.TuningMarkup)] CASPreferenceItemTuning = 3966303522`

CAS Preference Item Tuning

`[ResourceFileType(ResourceFileType.Binary)] CASPreset = 3936561885`

CAS Preset

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASStoriesAnswerTuning = 2163289367
```

CAS Stories Answer Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASStoriesQuestionTuning = 52718493
```

CAS Stories Question Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASStoriesTraitChooserTuning = 2376930633
```

CAS Stories Trait Chooser Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASThumbnailPartTuning = 2932986028
```

CAS Thumbnail Part Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASTuning = 3796587143
```

CAS Tuning

```
CMRF = 1700995767
```

CMRF

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CallToActionTuning = 4114068192
```

Call To Action Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CameraTuning = 306800208
```

Camera Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CareerEventTuning = 2487354146
```

Career Event Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CareerGigTuning = 3436908253
```

Career Gig Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CareerLevelTuning = 745582072
```

Career Level Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CareerTrackTuning = 1221024995
```

Career Track Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CareerTuning = 1939434475
```

Career Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CheatSheetTuning = 380861306
```

Cheat Sheet Tuning (the game's `CHEAT_SHEET`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClanTuning = 3737052837
```

Clan Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClanValueTuning = 2576273579
```

Clan Value Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClientTutorialTuning = 645072749
```

Client Tutorial Tuning

```
[ResourceFileType(ResourceFileType.Binary)] Clip = 1797309683
```

Clip

```
ClipHeader = 3158986820
```

Clip Header

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClubInteractionGroupTuning =
4195351092
```

Club Interaction Group Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClubSeedTuning = 794407991
```

Club Seed Tuning

```
ColorBlendedTerrain = 2934464740
```

Color Blended Terrain

```
ColorList = 2810271350
```

Color List

```
ColorTimelineData = 4250380503
```

Color Timeline Data

```
Column = 493744591
```

Column

```
[ResourceFileType(ResourceFileType.Binary)] CombinedTuning = 1659456824
```

Combined Binary Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ConditionalLayerTuning =
2441338001
```

Conditional Layer Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Credits = 1721014748
```

Credits

```
Cursor = 647463969
```

Cursor

```
[ResourceFileType(ResourceFileType.Binary)] CutoutInfoTable = 2177505808
```

Cutout Info Table

```
[ResourceFileType(ResourceFileType.DirectDrawSurface)] DSTImage = 11720834
```

DST Image

```
DecoTrim = 332336850
```

Deco Trim

```
[ResourceFileType(ResourceFileType.Binary)] DeformerMap = 3678658665
```

Deformer Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DetectiveClueTuning = 1400130038
```

Detective Clue Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DevelopmentalMilestoneTuning = 3307360148
```

Developmental Milestone Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DramaNodeTuning = 626258997
```

Drama Node Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DynastySeedTuning = 2214177389
```

Dynasty Seed Tuning (the game's `DYNASTY_SEED`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] EnsembleTuning = 3112702240
```

Ensemble Tuning

```
Fence = 68746794
```

Fence

```
Floor = 3036111561
```

Floor

```
FloorTrim = 2227319321
```

Floor Trim

```
[ResourceFileType(ResourceFileType.TuningMarkup)] FontConfiguration = 53690476
```

Font Configuration

```
[ResourceFileType(ResourceFileType.Binary)] Footprint = 3548561239
```

Footprint

```
Footprint3 = 3073696847
```

Footprint 3

```
Foundation = 799971390
```

Foundation

`FountainTrim = 3886917533`

Fountain Trim

`Frieze = 2690089244`

Frieze

`GPINI = 2249506521`

GPINI (the game's `GPINI`)

`[ResourceFileType(ResourceFileType.TuningMarkup)] GameRulesetTuning = 3779558936`

Game Ruleset Tuning

`GenericMTX = 2885921078`

Generic MTX

`[ResourceFileType(ResourceFileType.Binary)] Geometry = 22681673`

Geometry

`[ResourceFileType(ResourceFileType.TuningMarkup)] GetawayChallengeTuning = 3731146635`

Getaway Challenge Tuning (the game's `GETAWAY_CHALLENGE`)

`[ResourceFileType(ResourceFileType.TuningMarkup)] GuidanceTipTuning = 3567295165`

Guidance Tip Tuning

`HalfWall = 2438063804`

HalfWall

`HalfWallTrim = 2851789917`

Half Wall Trim (the game's `HALFWALL_TRIM`)

`[ResourceFileType(ResourceFileType.TuningMarkup)] HeadlineTuning = 4093714525`

Headline Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] HolidayDefinitionTuning = 238120813`

Holiday Definition Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HolidayTraditionTuning = 1070408838
```

Holiday Tradition Tuning

```
HouseholdDescription = 1923050575
```

Household Description

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HouseholdMilestoneTuning = 963831539
```

Household Milestone Tuning

```
HouseholdTemplate = 3015981296
```

Household Template

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HsvTweakerSettingsTuning = 3608008606
```

Hsv Tweaker Settings Tuning (the game's `HsvTweakerSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] InteractionTuning = 3900887599
```

Interaction Tuning

```
LRLEImage = 734023391
```

LRLE Image

```
Ladder = 3423141965
```

Ladder

```
Light = 62178845
```

Light

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LocomotionBuilder = 87703163
```

Locomotion Builder

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LocomotionConfig = 2600355829
```

Locomotion Config

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotDecorationPresetTuning = 3726571771
```

Lot Decoration Preset Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotDecorationTuning = 4264407467
```

Lot Decoration Tuning

```
[ResourceFileType(ResourceFileType.Binary)] LotDescription = 26488364
```

Lot Description

```
LotFootprintReference = 3221768598
```

Lot Footprint Reference

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)] LotPreviewThumbnail = 221481530
```

Lot Preview Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotTuning = 3632270694
```

Lot Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotTypeEventMap = 305120874
```

Lot Type Event Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LunarCycleTuning = 1430862616
```

Lunar Cycle Tuning

```
MTXBundle = 2377243942
```

MTX Bundle (the game's `MTX_BUNDLE`)

```
MTXCatalog = 1845434982
```

MTX Catalog

```
Magalog = 3105757403
```

Magalog

```
MagazineCollection = 1946487583
```

Magazine Collection

```
MaterialDefinition = 30467933
```

Material Definition

```
MaterialSet = 33659250
```

Material Set

```
MaxisWorldPipeline1 = 3773657385
```

Maxis World Pipeline 1

```
MaxisWorldPipeline2 = 4196775902
```

Maxis World Pipeline 2

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] MemorialThumbnail = 21
```

Memorial Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ModalMusicMapping = 444925637
```

Modal Music Mapping

```
Model = 23466547
```

Model

```
ModelCutout = 123169303
```

Model Cutout

```
ModelLOD = 30478132
```

Model LOD

```
ModularPiece = 2568481485
```

Modular Piece

```
ModularPieceCatalog = 2688883901
```

Modular Piece Catalog

```
[ResourceFileType(ResourceFileType.TuningMarkup)] MoodTuning = 3128647864
```

Mood Tuning

```
MusicData = 3254962032
```

Music Data

```
NameMap = 23462796
```

Name Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NarrativeTuning = 1047870521
```

Narrative Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NativeBuildBuyTuning = 2580268884
```

Native Build Buy Tuning (the game's `NativeBuildBuy`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NativeSeasonsWeatherTuning =
3579488305
```

Native Seasons Weather Tuning (the game's `Native_SeasonsWeather`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NotebookEntryTuning = 2567109238
```

Notebook Entry Tuning

```
ObjDef = 3625704905
```

Object Definition, distinct from ObjectDefinition (the game's `OBJDEF`)

```
[ResourceFileType(ResourceFileType.Binary)] ObjectCatalog = 832458525
```

Object Catalog

```
ObjectCatalogSet = 4283826444
```

Object Catalog Set

```
[ResourceFileType(ResourceFileType.Binary)] ObjectDefinition = 3235601127
```

Object Definition

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectModifiers = 3794908120
```

Object Modifiers

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectPartTuning = 1900520272
```

Object Part Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectStateTuning = 1526890910
```

Object State Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectTuning = 3055412916
```

Object Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectiveTuning = 6899006
```

Objective Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] OpenStreetDirectorTuning =
1265622724
```

Open Street Director Tuning

```
OpenTypeFont = 628714954
```

OpenType Font

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] PNG = 18461549
```

PNG

```
[ResourceFileType(ResourceFileType.DirectDrawSurfaceAsPortableNetworkGraphic)]
PNGImage = 796721156
```

PNG Image converted from DDS Image

```
[ResourceFileType(ResourceFileType.DirectDrawSurfaceAsPortableNetworkGraphic)]
PNGImage2 = 796721158
```

PNG Image converted from DDS Image 2

```
Path = 975082574
```

Path

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] PetBreedCoatPatternThumbnail = 2389771869
```

Pet Breed Coat Pattern Thumbnail

```
PetCoatBrush = 2642310488
```

Pet Coat Brush

```
PetCoatPattern = 3302993517
```

Pet Coat Pattern

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] PetFacePresetThumbnail = 3061216162
```

Pet Face Preset Thumbnail

```
PetPeltLayer = 649036600
```

Pet Pelt Layer

```
[ResourceFileType(ResourceFileType.TuningMarkup)] PieMenuCategoryTuning = 65657188
```

Pie Menu Category Tuning

```
Platform = 2579526276
```

Platform

```
Pond = 1342387004
```

Pond

```
PoolTrim = 2782919923
```

Pool Trim

```
[ResourceFileType(ResourceFileType.TuningMarkup)] PostureTuning = 2909789983
```

Posture Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] PubertyTuning = 1209600077
```

Puberty Tuning

```
QueryableWorldMaskManifest = 1316074726
```

Queryable World Mask Manifest

```
[ResourceFileType(ResourceFileType.DirectDrawSurface)] RLE2Image = 877907861
```

RLE 2 Image

```
[ResourceFileType(ResourceFileType.DirectDrawSurface)] RLESImage = 3129306232
```

RLES Image

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RabbitHoleTuning = 2976568058
```

Rabbit Hole Tuning

```
Railing = 471658999
```

Railing

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RecipeTuning = 3952605219
```

Recipe Tuning

```
RecordBundle = 1051371644
```

Record Bundle

```
[ResourceFileType(ResourceFileType.Binary)] RegionDescription = 3596464121
```

Region Description

```
[ResourceFileType(ResourceFileType.Binary)] RegionMap = 2887187436
```

Region Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RegionSortTuning = 825849413
```

Region Sort Tuning (the game's `RegionSort`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RegionTuning = 1374134669
```

Region Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RelationshipBitTuning = 151314192
```

Relationship Bit Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RelationshipLockTuning = 2922702451
```

Relationship Lock Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererCensorTuning = 1166000178
```

Renderer Censor Tuning (the game's `Renderer_CensorTuning`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererFadeTuning = 250315322
```

Renderer Fade Tuning (the game's `Renderer_FadeTuning`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalDOFSettingsTuning = 1303913758
```

Renderer Global DOF Settings Tuning (the game's `Renderer_GlobalDofSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalHighlightSettingsTuning = 738311189
```

Renderer Global Highlight Settings Tuning (the game's `Renderer_GlobalHighlightSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalLightSettingsTuning = 2249419429
```

Renderer Global Light Settings Tuning (the game's `Renderer_GlobalLightSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalSSAOSettingsTuning = 4181409999
```

Renderer Global SSAO Settings Tuning (the game's `Renderer_GlobalSsaoSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalShadowSettingsTuning = 934910449
```

Renderer Global Shadow Settings Tuning (the game's `Renderer_GlobalShadowSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalVolLightScatteringSettingsTuning = 3765102174
```

Renderer Global Vol Light Scattering Settings Tuning (the game's `Renderer_GlobalVolLightScatteringSettings`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RewardTuning = 1873057832
```

Reward Tuning

```
Rig = 2393838558
```

Rig

```
RoadDefinition = 2422433294
```

Road Definition

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RoleStateTuning = 239932923
```

Role State Tuning

```
Roof = 2448276798
```

Roof

```
RoofPattern = 4058889606
```

Roof Pattern

```
RoofTrim = 2956008719
```

Roof Trim

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RoyaltyTuning = 938421991
```

Royalty Tuning

```
S4sMergedPackageManifest = 2142678410
```

Sims 4 Studio Merged Package Manifest

```
[ResourceFileType(ResourceFileType.JointPhotographicExpertsGroupImage)]
SaveGameCustomTexture = 3901600607
```

Save Game Custom Texture

```
[ResourceFileType(ResourceFileType.Binary)] SaveGameData = 13
```

Save Game Data

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameHouseholdThumbnail = 20
```

Save Game Household Thumbnail

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotLevelThumbnail = 18
```

Save Game Lot Level Thumbnail

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotLevelThumbnailMask = 19
```

Save Game Lot Level Thumbnail Mask

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotThumbnail1 = 15
```

Save Game Lot Thumbnail 1

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotThumbnail2 = 16
```

Save Game Lot Thumbnail 2

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameSimCustomTexture = 4175512954
```

Save Game Sim Custom Texture

```
[ResourceFileType(ResourceFileType.Binary)] SaveSpecificRelationalDataStorage =
1382263428
```

Save-Specific Relational Data Storage

This is a ZIP archive's octet stream set to store, not any level of compression. Use DBPF ZLIB compression. Each entry should be a SQLite database file using the `.sqlite` extension, the name of which is a valid UUID.

```
ScaleFormGFX = 1659684250
```

ScaleForm GFX

```
Sculpt = 2635774068
```

Sculpt

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SeasonTuning = 3381515358
```

Season Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] ServiceNPCTuning = 2629964386

Service NPC Tuning

[ResourceFileType(ResourceFileType.Binary)] ShaderPrecomp = 670978917

Game Shaders for Windows

[ResourceFileType(ResourceFileType.TuningMarkup)] SicknessTuning = 3288062174

Sickness Tuning

[ResourceFileType(ResourceFileType.Binary)] SimData = 1415235194

Sim Data

[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)] SimFeaturedOutfitThumbnail = 3449676359

Sim Featured Outfit Thumbnail

[ResourceFileType(ResourceFileType.TuningMarkup)] SimFilterTuning = 1846401695

Sim Filter Tuning

SimHotspotControl = 2333671278

Sim Hotspot Control

[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)] SimHouseholdThumbnail = 1003770887

Sim Household Thumbnail

SimInfo = 39769844

Sim Info

[ResourceFileType(ResourceFileType.TuningMarkup)] SimInfoFixupTuning = 3797424274

Sim Info Fixup Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] SimLightingTuning = 2648399834

Sim Lighting Tuning

SimModifier = 3321263678

Sim Modifier

```
SimPreset = 273810874
```

Sim Preset

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SimPresetThumbnail = 2626836499
```

Sim Preset Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SimTemplateTuning = 212125579
```

Sim Template Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SituationGoalSetTuning =
2649944562
```

Situation Goal Set Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SituationGoalTuning = 1502554343
```

Situation Goal Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SituationJobTuning = 2617738591
```

Situation Job Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SituationTuning = 4223905515
```

Situation Tuning

```
[ResourceFileType(ResourceFileType.Binary)] Skintone = 55867754
```

Skintone

```
[ResourceFileType(ResourceFileType.Binary)] SkyBoxTextureData = 1906592201
```

Sky Box Texture Data

```
[ResourceFileType(ResourceFileType.Binary)] Slot = 3540272417
```

Slot

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SlotTypeSetTuning = 1058419973
```

Slot Type Set Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SlotTypeTuning = 1772477092
```

Slot Type Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SmallBusinessBinaryTuning =
91497693
```

Small Business Binary Tuning (the game's `SMALL_BUSINESS_BINARY`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SmallBusinessSeedTuning =
2989400653
```

Small Business Seed Tuning (the game's `SMALL_BUSINESS_SEED`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SnippetTuning = 2113017500
```

Snippet Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SocialGroupTuning = 776446212
```

Social Group Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SoundMixProperties = 1091959253
```

Sound Mix Properties

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SoundModifierMapping = 2776023783
```

Sound Modifier Mapping

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SoundProperties = 455450660
```

Sound Properties

```
Spandrel = 1057772186
```

Spandrel

```
Spawner = 1220708729
```

Spawner

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SpellTuning = 523506649
```

Spell Tuning

```
Stairs = 2585840924
```

Stairs

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StaticCommodityTuning =
1359443523
```

Static Commodity Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StatisticTuning = 865846717
```

Statistic Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StoryArcTuning = 1613438381
```

Story Arc Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StoryChapterTuning = 1250314810
```

Story Chapter Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StrategySetTuning = 1646578134
```

Strategy Set Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StreetTuning = 4142189312
```

Street Tuning

```
[ResourceFileType(ResourceFileType.Binary)] StringTable = 570775514
```

String Table

```
Style = 2673671952
```

Style

```
StyledLook = 1908258978
```

Styled Look

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SubrootTuning = 3086978965
```

Subroot Tuning

```
SyncPointSchema = 757559827
```

Sync Point Schema

[ResourceFileType(ResourceFileType.TuningMarkup)] TagCategoriesMetadataTuning = 470996056

Tag Categories Metadata Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] TagSetTuning = 1228493570

Tag Set Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] TagTraitGroupMetadataTuning = 2302558876

Tag Trait Group Metadata Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] TagTuning = 3708124620

Tag Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] TagsMetadataTuning = 2646182397

Tags Metadata Tuning

[ResourceFileType(ResourceFileType.TuningMarkup)] TelemetryMemoryUsageTelemetrySettingsTuning = 979055765

Telemetry Memory Usage Telemetry Settings Tuning (the game's `Telemetry_MemoryUsageTelemetrySettings`)

[ResourceFileType(ResourceFileType.TuningMarkup)] TelemetryTuning = 3268060742

Telemetry Tuning (the game's `Telemetry_Tuning`)

[ResourceFileType(ResourceFileType.TuningMarkup)] TemplateChooserTuning = 1220728301

Template Chooser Tuning

[ResourceFileType(ResourceFileType.Binary)] TerrainBlendMap = 1032205008

Terrain Blend Map

[ResourceFileType(ResourceFileType.Binary)] TerrainData = 2422433293

Terrain Data

[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)] TerrainHeightMap = 718378482

Terrain Height Map

[ResourceFileType(ResourceFileType.Binary)] TerrainKDTree = 54209382

Terrain KD Tree

[ResourceFileType(ResourceFileType.Binary)] TerrainMesh = 2922985887

Terrain Mesh

TerrainPaint = 3955994988

Terrain Paint

TerrainSizeInfo = 2422361115

Terrain Size Info

TerrainTool = 338149641

Terrain Tool

[ResourceFileType(ResourceFileType.TuningMarkup)] TestBasedScoreTuning = 1332976878

Test Based Score Tuning

TheSimsResourceWorkshopProprietary = 20281636

A proprietary resource used by The Sims Resource Workshop

[ResourceFileType(ResourceFileType.TuningMarkup)] ThriftStoreTuning = 3841838852

Thrift Store Tuning

[ResourceFileType(ResourceFileType.Binary)] ThumbnailCache = 3107625237

Thumbnail Cache

ThumbnailExtra1 = 382364612

Thumbnail Extra 1

ThumbnailExtra2 = 2953939989

Thumbnail Extra 2

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ThumbnailTuning = 3627586231
```

Thumbnail Tuning (the game's `Thumbnail`)

```
TimelineEvents = 111944783
```

Timeline Events

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TopicTuning = 1938713686
```

Topic Tuning

```
TrackMask = 53633251
```

Track Mask

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TraitTuning = 3412057543
```

Trait Tuning

```
TrayItem = 713711138
```

Tray Item

```
[ResourceFileType(ResourceFileType.Binary)] Trim = 1992095756
```

Trim

```
TrueTypeFont = 661431481
```

TrueType Font

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Tuning = 62078431
```

Tuning

```
[ResourceFileType(ResourceFileType.Binary)] TuningDependenciesCache = 879540542
```

Tuning Dependencies Cache (the game's `TD_CACHE`)

```
TuningDescription = 2519486516
```

Tuning Description (the game's `TUNING_DESCRIPTION`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TutorialTipTuning = 2410930353
```

Tutorial Tip Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TutorialTuning = 3762955427
```

Tutorial Tuning

```
UIControlEventMap = 3185058337
```

UI Control Event Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UIEventModeMapping = 2581168265
```

UI Event Mode Mapping

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityCourseDataTuning =
689745854
```

University Course Data Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityCourseMajorTuning =
660124491
```

University Course Major Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityCourseScheduleTuning =
2559322869
```

University Course Schedule Tuning (the game's `UNIVERSITY_COURSE_SCHEDULE`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityTuning = 3646477745
```

University Tuning

```
UnknownWorld2 = 398508673
```

Unknown World 2

```
UnknownWorld3 = 2528165143
```

Unknown World 3

```
UnknownWorld5 = 3162565977
```

Unknown World 5

```
Unspecified = 0
```

Unspecified

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UserInterfaceInfoTuning = 3099531875
```

User Interface Info Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VenueTuning = 3871070174
```

Venue Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VideoGlobalTuning = 3731542991
```

Video Global Tuning (the game's `Video_GlobalTuning`)

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VideoPlaylistTuning = 3848202955
```

Video Playlist Tuning (the game's `Video_Playlist`)

```
VisualEffects = 454631497
```

Visual Effects

```
VisualEffectsInstanceMap = 454631498
```

Visual Effects Instance Map

```
VisualEffectsMerged = 3931183280
```

Visual Effects Merged

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VoiceEffect = 1942696642
```

Voice Effect

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VoicePlugin = 3313685243
```

Voice Plugin

```
WMRFReference = 551031958
```

WMRF Reference

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WalkByTuning = 1070998590
```

Walk By Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Walkstyle = 666901909
```

Walkstyle

```
Wall = 3589339425
```

Wall

```
WallTrim = 4264756878
```

Wall Trim

```
WaterManifest = 386491005
```

Water Manifest

```
WaterMask = 1207820220
```

Water Mask

```
WaterMaskList = 2763687493
```

Water Mask List

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WeatherEventTuning = 1476851130
```

Weather Event Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WeatherForecastTuning =
1233072753
```

Weather Forecast Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WhimTuning = 1956251190
```

Whim Tuning

```
WindowSet = 2834806039
```

Window Set

```
WorldCameraInfo = 2301381514
```

World Camera Info

```
WorldCameraMesh = 4211986434
```

World Camera Mesh

```
[ResourceFileType(ResourceFileType.Binary)] WorldConditionalData = 3494881377
```

World Conditional Data

```
[ResourceFileType(ResourceFileType.Binary)] WorldData = 2164920365
```

World Data

```
[ResourceFileType(ResourceFileType.Binary)] WorldDescription = 2793466443
```

World Description

```
WorldFileHeader = 4033034633
```

World File Header

```
WorldLandingStrip = 1332898750
```

World Landing Strip

```
WorldLightsInfo = 418629235
```

World Lights Info

```
WorldLotArchitecture = 311764532
```

World Lot Architecture

```
WorldLotObjects = 2438369240
```

World Lot Objects

```
WorldLotParameterInfo = 1006173574
```

World Lot ParameterInfo

```
WorldManifest = 2026421476
```

World Manifest

```
[ResourceFileType(ResourceFileType.Binary)] WorldMap = 482361971
```

World Map

```
[ResourceFileType(ResourceFileType.Binary)] WorldObjectData = 4239499748
```

World Object Data

```
WorldOffLotMesh = 170032079
```

World Off Lot Mesh

```
WorldRoadPolys = 1866496362
```

World Road Polys

```
[ResourceFileType(ResourceFileType.Binary)] WorldTimelineColor = 422580512
```

World Timeline Color

```
WorldVisualEffectsInfo = 1541576451
```

World Visual Effects Info

```
WorldWaterManifest = 356327961
```

World Water Manifest

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] WorldmapLotThumbnail = 2717855684
```

Worldmap Lot Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ZoneDirectorTuning = 4183335058
```

Zone Director Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ZoneModifierTuning = 1008568217
```

Zone Modifier Tuning

```
ZoneObjectData = 6
```

Zone Object Data

Class SmartSimUtilities

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Provides utility methods specific to The Sims 4

```
public static class SmartSimUtilities
```

Inheritance

object ← SmartSimUtilities

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetResourceKeyForLocale(ResourceKey, CultureInfo)

Gets the ResourceKey for this string table resource with the specified `locale`

```
public static ResourceKey GetResourceKeyForLocale(this ResourceKey key, CultureInfo locale)
```

Parameters

`key` ResourceKey

`locale` CultureInfo

Returns

ResourceKey

Exceptions

ArgumentOutOfRangeException

The locale is not supported by The Sims 4

GetStringTableLocale(ResourceKey)

Gets the locale of a string table resource in The Sims 4 based on its `key`

```
public static CultureInfo GetStringTableLocale(this ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

CultureInfo

Exceptions

ArgumentOutOfRangeException

The high order byte of the instance does not correspond to a locale supported by The Sims 4

GetStringTableResourceKey(CultureInfo, uint, ulong)

Gets the ResourceKey of a string table resource in The Sims 4 based on its `locale`, `group`, and `fullInstance`

```
public static ResourceKey GetStringTableResourceKey(this CultureInfo locale, uint group,
ulong fullInstance)
```

Parameters

`locale` CultureInfo

`group` uint

`fullInstance` ulong

Returns

ResourceKey

Exceptions

ArgumentOutOfRangeException

The locale is not supported by The Sims 4

Namespace LlamaLogic.Packages.Cryptography

Classes

Fnv24

Represents the 24-bit implementation of the Fowler–Noll–Vo hash algorithm

Fnv32

Represents the 32-bit implementation of the Fowler–Noll–Vo hash algorithm

Fnv56

Represents the 56-bit implementation of the Fowler–Noll–Vo hash algorithm

Fnv64

Represents the 64-bit implementation of the Fowler–Noll–Vo hash algorithm

FnvHash

Represents the base class from which this library's implementations of the Fowler–Noll–Vo hash algorithms derive

Class Fnv24

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 24-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public sealed class Fnv24 : Fnv32, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv32 ← Fnv24

Implements

ICryptoTransform, IDisposable

Inherited Members

Fnv32.HashSize , FnvHash.ComputeHash(string) , FnvHash.Initialize() , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly uint EmptyStringHash
```

Field Value

uint

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static uint GetHash(string? text)
```

Parameters

`text` string

Returns

uint

SetHighBit(uint)

Sets the high bit of a 24-bit Fowler–Noll–Vo hash

```
public static uint SetHighBit(uint hash)
```

Parameters

`hash` uint

Returns

uint

Class Fnv32

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 32-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public class Fnv32 : FnvHash, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv32

Implements

ICryptoTransform, IDisposable

Derived

Fnv24

Inherited Members

FnvHash.hash , FnvHash.ComputeHash(string) , FnvHash.HashCore(byte[], int, int) , FnvHash.HashFinal() , FnvHash.Initialize() , HashAlgorithm.HashSizeValue , HashAlgorithm.HashValue , HashAlgorithm.State , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.Dispose(bool) , HashAlgorithm.HashCore(ReadOnlySpan<byte>) , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.TryHashFinal(Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Fnv32()

Initializes a new instance of the Fnv32 class

```
public Fnv32()
```

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly uint EmptyStringHash
```

Field Value

uint

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

HashSize

Gets the size, in bits, of the computed hash code.

```
public override int HashSize { get; }
```

Property Value

int

The size, in bits, of the computed hash code.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static uint GetHash(string? text)
```

Parameters

`text` string

Returns

uint

SetHighBit(uint)

Sets the high bit of a 32-bit Fowler–Noll–Vo hash

```
public static uint SetHighBit(uint hash)
```

Parameters

`hash` uint

Returns

uint

Class Fnv56

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 56-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public sealed class Fnv56 : Fnv64, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv64 ← Fnv56

Implements

ICryptoTransform, IDisposable

Inherited Members

Fnv64.HashSize , FnvHash.ComputeHash(string) , FnvHash.Initialize() , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly ulong EmptyStringHash
```

Field Value

ulong

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static ulong GetHash(string? text)
```

Parameters

`text` string

Returns

ulong

SetHighBit(ulong)

Sets the high bit of a 56-bit Fowler–Noll–Vo hash

```
public static ulong SetHighBit(ulong hash)
```

Parameters

`hash` ulong

Returns

ulong

Class Fnv64

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 64-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public class Fnv64 : FnvHash, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv64

Implements

ICryptoTransform, IDisposable

Derived

Fnv56

Inherited Members

FnvHash.hash , FnvHash.ComputeHash(string) , FnvHash.HashCore(byte[], int, int) , FnvHash.HashFinal() , FnvHash.Initialize() , HashAlgorithm.HashSizeValue , HashAlgorithm.HashValue , HashAlgorithm.State , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.Dispose(bool) , HashAlgorithm.HashCore(ReadOnlySpan<byte>) , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.TryHashFinal(Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Fnv64()

Initializes a new instance of the Fnv64 class

```
public Fnv64()
```

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly ulong EmptyStringHash
```

Field Value

ulong

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

HashSize

Gets the size, in bits, of the computed hash code.

```
public override int HashSize { get; }
```

Property Value

int

The size, in bits, of the computed hash code.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static ulong GetHash(string? text)
```

Parameters

`text` string

Returns

ulong

SetHighBit(ulong)

Sets the high bit of a 64-bit Fowler–Noll–Vo hash

```
public static ulong SetHighBit(ulong hash)
```

Parameters

`hash` ulong

Returns

ulong

Class FnvHash

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the base class from which this library's implementations of the Fowler–Noll–Vo hash algorithms derive

```
public abstract class FnvHash : HashAlgorithm, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash

Implements

ICryptoTransform, IDisposable

Derived

Fnv32, Fnv64

Inherited Members

HashAlgorithm.HashSizeValue , HashAlgorithm.HashValue , HashAlgorithm.State , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.Dispose(bool) , HashAlgorithm.HashCore(ReadOnlySpan<byte>) , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.TryHashFinal(Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.Hash , HashAlgorithm.HashSize , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FnvHash(ulong, ulong)

Represents the base class from which this library's implementations of the Fowler–Noll–Vo hash algorithms derive

```
protected FnvHash(ulong prime, ulong offset)
```

Parameters

`prime` ulong

`offset` ulong

Fields

hash

The hash aggregate

```
protected ulong hash
```

Field Value

ulong

Methods

ComputeHash(string?)

Computes the hash value for the specified string

```
[SuppressMessage("Globalization", "CA1308: Normalize strings to uppercase", Justification =
"It's not our fault Maxis did this wrong, Code Analyzer.")]
public byte[] ComputeHash(string? value)
```

Parameters

`value` string

Returns

byte[]

HashCore(byte[], int, int)

When overridden in a derived class, routes data written to the object into the hash algorithm for computing the hash.

```
protected override void HashCore(byte[] array, int ibStart, int cbSize)
```

Parameters

`array` byte[]

The input to compute the hash code for.

`ibStart` int

The offset into the byte array from which to begin using data.

`cbSize` int

The number of bytes in the byte array to use as data.

HashFinal()

When overridden in a derived class, finalizes the hash computation after the last data is processed by the cryptographic hash algorithm.

```
protected override byte[] HashFinal()
```

Returns

byte[]

The computed hash code.

Initialize()

Resets the hash algorithm to its initial state.

```
public override void Initialize()
```

Namespace LlamaLogic.Packages.Formats

Classes

DirectDrawSurface

Provides methods for dealing with diffuse surface textures in Direct Draw Surfaces

Ts4TranslucentJointPhotographicExpertsGroupImage

Provides methods for dealing with Maxis Translucent JPEGs

Class DirectDrawSurface

Namespace: LlamaLogic.Packages.Formats

Assembly: LlamaLogic.Packages.dll

Provides methods for dealing with diffuse surface textures in Direct Draw Surfaces

```
public static class DirectDrawSurface
```

Inheritance

object ← DirectDrawSurface

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Adapted from S4S by Andrew

Methods

GetDstDataFromPngData(ReadOnlyMemory<byte>)

Convert a Portable Network Graphic to a shuffled DirectDraw Surface compatible with The Sims 4

```
public static ReadOnlyMemory<byte> GetDstDataFromPngData(ReadOnlyMemory<byte> pngData)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data of the Portable Network Graphic

Returns

ReadOnlyMemory<byte>

Exceptions

InvalidDataException

`pngData` is not a PNG, could not be decoded, or has more than 67,108,864 (8,192 by 8,192) pixels

GetDstDataFromPngDataAsync(ReadOnlyMemory<byte>, CancellationToken)

Convert a Portable Network Graphic to a shuffled DirectDraw Surface compatible with The Sims 4, asynchronously

```
public static Task<ReadOnlyMemory<byte>> GetDstDataFromPngDataAsync(ReadOnlyMemory<byte>
pngData, CancellationToken cancellationToken = default)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data of the Portable Network Graphic

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

Exceptions

InvalidDataException

`pngData` is not a PNG, could not be decoded, or has more than 67,108,864 (8,192 by 8,192) pixels

GetPngDataFromDdsData(ReadOnlyMemory<byte>)

Convert a DirectDraw Surface to a Portable Network Graphic

```
public static ReadOnlyMemory<byte> GetPngDataFromDdsData(ReadOnlyMemory<byte> ddsData)
```

Parameters

`ddsData` ReadOnlyMemory<byte>

The data of a DirectDraw Surface

Returns

ReadOnlyMemory<byte>

Exceptions

InvalidDataException

The surface has no pixels or more than 67,108,864 (8,192 by 8,192) pixels, or declares more than 32 mipmap levels

GetPngDataFromDdsDataAsync(ReadOnlyMemory<byte>, CancellationToken)

Convert a DirectDraw Surface to a Portable Network Graphic, asynchronously

```
public static Task<ReadOnlyMemory<byte>> GetPngDataFromDdsDataAsync(ReadOnlyMemory<byte>
ddsData, CancellationToken cancellationToken = default)
```

Parameters

`ddsData` ReadOnlyMemory<byte>

The data of a DirectDraw Surface

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

Exceptions

InvalidDataException

The surface has no pixels or more than 67,108,864 (8,192 by 8,192) pixels, or declares more than 32 mipmap levels

Class Ts4TranslucentJointPhotographicExpertsGroupImage

Namespace: LlamaLogic.Packages.Formats

Assembly: LlamaLogic.Packages.dll

Provides methods for dealing with Maxis Translucent JPEGs

```
public static class Ts4TranslucentJointPhotographicExpertsGroupImage
```

Inheritance

object ← Ts4TranslucentJointPhotographicExpertsGroupImage

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConvertPngToTranslucentJpeg(ReadOnlyMemory<byte>)

Converts the data for a PNG the data for a Maxis Translucent JPEG

```
public static ReadOnlyMemory<byte> ConvertPngToTranslucentJpeg(ReadOnlyMemory<byte> pngData)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data for a PNG

Returns

ReadOnlyMemory<byte>

Exceptions

InvalidDataException

`pngData` is not a PNG, could not be decoded, or has more than 67,108,864 (8,192 by 8,192) pixels, or its alpha channel compresses to more than 65,525 bytes

ConvertPngToTranslucentJpegAsync(ReadOnlyMemory<byte>, CancellationToken)

Converts the data for a PNG the data for a Maxis Translucent JPEG asynhronously

```
public static Task<ReadOnlyMemory<byte>>
ConvertPngToTranslucentJpegAsync(ReadOnlyMemory<byte> pngData,
CancellationToken cancellationToken)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data for a PNG

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

Exceptions

InvalidDataException

`pngData` is not a PNG, could not be decoded, or has more than 67,108,864 (8,192 by 8,192) pixels, or its alpha channel compresses to more than 65,525 bytes

ConvertTranslucentJpegToPng(ReadOnlyMemory<byte>)

Converts the data for a Maxis Translucent JPEG the data for a PNG

```
public static ReadOnlyMemory<byte> ConvertTranslucentJpegToPng(ReadOnlyMemory<byte>
translucentJpegData)
```

Parameters

`translucentJpegData` ReadOnlyMemory<byte>

The data for a Maxis Translucent JPEG

Returns

ReadOnlyMemory<byte>

Exceptions

FormatException

`translucentJpegData` is not a JPEG

InvalidDataException

The ALFA PNG APP0 marker is missing or malformed, the JPEG or the PNG could not be decoded or has more than 67,108,864 (8,192 by 8,192) pixels, or their dimensions differ

ConvertTranslucentJpegToPngAsync(ReadOnlyMemory<byte>, CancellationToken)

Converts the data for a Maxis Translucent JPEG the data for a PNG asynhronously

```
public static Task<ReadOnlyMemory<byte>>
ConvertTranslucentJpegToPngAsync(ReadOnlyMemory<byte> translucentJpegData, CancellationToken
cancellationToken = default)
```

Parameters

`translucentJpegData` ReadOnlyMemory<byte>

The data for a Maxis Translucent JPEG

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

Exceptions

FormatException

`translucentJpegData` is not a JPEG

InvalidDataException

The ALFA PNG APP0 marker is missing or malformed, the JPEG or the PNG could not be decoded or has more than 67,108,864 (8,192 by 8,192) pixels, or their dimensions differ

Namespace LlamaLogic.Packages.Models

Classes

Model

Represents a model for a raw resource

StringTableModel

Represents a StringTable resource (🔒)

UnexpectedFormatDecodingException

Represents an exception thrown by an IModel<TSelf> when it attempts to decode a resource that is not in the expected format

Interfaces

IModel

Represents a model for a resource

IModel<TSelf>

Represents a model for a raw resource

Interface IModel

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a model for a resource

```
public interface IModel
```

Properties

ResourceName

Gets the name of this resource if it has one

```
string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static abstract ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static abstract string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static abstract Task<string?> GetNameAsync(Stream stream, CancellationToken
cancellationToken = default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Interface IModel<TSelf>

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a model for a raw resource

```
public interface IModel<TSelf> : IModel where TSelf : IModel<TSelf>
```

Type Parameters

`TSelf`

Inherited Members
IModel.SupportedTypes , IModel.GetName(Stream) , IModel.GetNameAsync(Stream, CancellationToken) , IModel.ResourceName , IModel.Encode()

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔁 🏃)

```
public static abstract TSelf Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

TSelf

Class Model

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a model for a raw resource

```
public abstract class Model : IModel
```

Inheritance
object ← Model

Implements
IModel

Derived
CombinedTuningModel, GlobalModsManifestModel, ModFileManifestModel, SimDataModel, StringTableModel

Inherited Members
object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ResourceName

Gets the name of this resource if it has one

```
public abstract string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

Decode(ResourceType, ReadOnlyMemory<byte>)

Decodes a resource of the specified `type` with the model GetModelType(ResourceType) gives for it

```
public static Model Decode(ResourceType type, ReadOnlyMemory<byte> data)
```

Parameters

`type` ResourceType

`data` ReadOnlyMemory<byte>

Returns

Model

Exceptions

NotSupportedException

No model decodes resources of the specified `type`

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public abstract ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

GetModelType(ResourceType)

Gets the model that decodes every resource of the specified `type`, or null if there is none

```
public static Type? GetModelType(ResourceType type)
```

Parameters

`type` ResourceType

Returns

Type

Remarks

Models that apply only to some resources of a type, such as ModFileManifestModel for SnippetTuning, are not given.

GetName(ResourceType, Stream)

Gets the name of a resource from its raw data, if it has one

```
public static string? GetName(ResourceType type, Stream stream)
```

Parameters

`type` ResourceType

`stream` Stream

Returns

string

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(ResourceType, Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one

```
public static Task<string?> GetNameAsync(ResourceType type, Stream stream, CancellationToken
cancellationToken = default)
```

Parameters

`type` ResourceType

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task <string>

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
```

```
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Class StringTableModel

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a StringTable resource (🔒)

```
public class StringTableModel : Model, IModel<StringTableModel>, IModel
```

Inheritance

object ← Model ← StringTableModel

Implements

IModel<StringTableModel>, IModel

Inherited Members

Model.Decode(ResourceType, ReadOnlyMemory<byte>) , Model.GetModelType(ResourceType) , Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Count

Gets the number of entries in the string table

```
public int Count { get; }
```

Property Value

int

this[uint]

Gets/sets the string associated with a `keyHash`

```
public string this[uint keyHash] { get; set; }
```

Parameters

`keyHash` uint

Property Value

string

Exceptions

ArgumentNullException

The value is null

ArgumentException

The value is longer than 65,535 bytes in UTF-8

KeyHashes

Gets the key hashes in the string table

```
public IReadOnlyList<uint> KeyHashes { get; }
```

Property Value

IReadOnlyList<uint>

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet <ResourceType>

Methods

AddNew(string)

Adds `string` to the string table, returning its key hash

```
public uint AddNew(string @string)
```

Parameters

`string` string

Returns

uint

Exceptions

ArgumentNullException

`string` is null

ArgumentException

`string` is longer than 65,535 bytes in UTF-8, or the key hash generated for it already existed in the string table

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔁 🏃)

```
public static StringTableModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

StringTableModel

Exceptions

InvalidDataException

> The data is not a well-formed string table: its file identifier or version is not the expected one, its entries do not fill it exactly, its total string length disagrees with its entries, or a string is not valid UTF-8

Delete(uint)

Deletes the string associated with a `keyHash`, returning true if the key hash existed in the string table

```
public bool Delete(uint keyHash)
```

Parameters

`keyHash` uint

Returns

bool

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

Get(uint)

Gets the string associated with a `keyHash`

```
public string Get(uint keyHash)
```

Parameters

`keyHash` uint

Returns

string

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Set(uint, string)

Sets the `string` associated with a `keyHash`, returning true if the key hash did not already exist in the string table

```
public bool Set(uint keyHash, string @string)
```

Parameters

`keyHash` uint

`string` string

Returns

bool

Exceptions

ArgumentNullException

`string` is null

ArgumentException

`string` is longer than 65,535 bytes in UTF-8

Class UnexpectedFormatDecodingException

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents an exception thrown by an IModel<TSelf> when it attempts to decode a resource that is not in the expected format

```
public sealed class UnexpectedFormatDecodingException : Exception, ISerializable
```

Inheritance

object ← Exception ← UnexpectedFormatDecodingException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.ReferenceEquals(object, object)

Constructors

UnexpectedFormatDecodingException()

Initializes a new UnexpectedFormatDecodingException

```
public UnexpectedFormatDecodingException()
```

UnexpectedFormatDecodingException(Exception?)

Initializes a new UnexpectedFormatDecodingException with the specified `innerException`

```
public UnexpectedFormatDecodingException(Exception? innerException)
```

Parameters

`innerException` Exception

UnexpectedFormatDecodingException(string)

Initializes a new UnexpectedFormatDecodingException with the specified `message`

```
public UnexpectedFormatDecodingException(string message)
```

Parameters

`message` string

UnexpectedFormatDecodingException(string, Exception?)

Initializes a new UnexpectedFormatDecodingException with the specified `message` and `innerException`

```
public UnexpectedFormatDecodingException(string message, Exception? innerException)
```

Parameters

`message` string

`innerException` Exception

Namespace LlamaLogic.Packages.Models.CombinedTuning

Classes

CombinedTuningModel

Represents a CombinedTuning resource (🔓)

CombinedTuningModelTuning

Represents an individual tuning document from a CombinedTuningModel (🔓)

Class CombinedTuningModel

Namespace: LlamaLogic.Packages.Models.CombinedTuning

Assembly: LlamaLogic.Packages.dll

Represents a CombinedTuning resource (🔓)

```
public sealed class CombinedTuningModel : Model, IModel<CombinedTuningModel>, IModel
```

Inheritance

object ← Model ← CombinedTuningModel

Implements

IModel<CombinedTuningModel>, IModel

Inherited Members

Model.Decode(ResourceType, ReadOnlyMemory<byte>) , Model.GetModelType(ResourceType) , Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Maxis ships the game's tuning merged into resources of this type, encoded either as a binary packed XML document (a `DATA` file) or as XML text. Decode(ReadOnlyMemory<byte>) accepts both and presents the merged document as Document. Encode() always produces XML text, written the way Maxis writes it. GetTunings() produces the individual tuning documents the merged document contains.

Constructors

CombinedTuningModel(XDocument)

Initializes a new CombinedTuningModel for the specified `document`

```
public CombinedTuningModel(XDocument document)
```

Parameters

`document` XDocument

The merged document, the root element of which is `<combined>`

Properties

Document

Gets the merged document, the root element of which is `<combined>`

```
public XDocument Document { get; }
```

Property Value

XDocument

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔄 🏃)

```
public static CombinedTuningModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

CombinedTuningModel

Exceptions

InvalidDataException

The data is neither a well-formed binary nor a well-formed XML combined tuning document

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

Exceptions

InvalidOperationException

Document has no root element or contains a node that XML combined tuning cannot express

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task <string>

GetTunings()

Gets the individual tuning documents in Document, in document order, with each reference to a shared node (`<r x="…">`) replaced by a copy of that node (💤)

```
public IEnumerable<CombinedTuningModelTuning> GetTunings()
```

Returns

IEnumerable<CombinedTuningModelTuning>

Remarks

The documents are produced as the sequence is enumerated, and are independent of Document and of each other. Modifying Document while enumerating has undefined results.

Exceptions

InvalidDataException

Document is not a well-formed merged document: it lacks the shared node group, a reference names a node that does not exist or contains itself, or a tuning document lacks a valid instance

Class CombinedTuningModelTuning

Namespace: LlamaLogic.Packages.Models.CombinedTuning

Assembly: LlamaLogic.Packages.dll

Represents an individual tuning document from a CombinedTuningModel (🔓)

```
public sealed class CombinedTuningModelTuning
```

Inheritance

object ← CombinedTuningModelTuning

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Document

Gets the tuning document, as it would appear in a TuningMarkup resource

```
public XDocument Document { get; }
```

Property Value

XDocument

Instance

Gets the instance of the tuning (its `s` attribute)

```
public ulong Instance { get; }
```

Property Value

ulong

Name

Gets the name of the tuning (its `n` attribute), if it has one

```
public string? Name { get; }
```

Property Value

string

ResourceType

Gets the type of the tuning's resource, derived from TypeName the way the game does

```
public ResourceType ResourceType { get; }
```

Property Value

ResourceType

Remarks

The game's type is the FNV-1 32-bit hash of the type name, except for `tun`, `relbit` and `notebook_entry`, which it assigns Tuning, RelationshipBitTuning and NotebookEntryTuning. The value may not be a defined ResourceType member when the game introduces a type this library does not yet name.

TypeName

Gets the name the merged document gives the tuning's type (the `n` attribute of its `<R>` group, e.g. `buff` or `Renderer_GlobalShadowSettings`)

```
public string TypeName { get; }
```

Property Value

string

Namespace LlamaLogic.Packages.Models.GlobalManifest

Classes

GlobalModsManifestModel

Represents the global mods manifest SnippetTuning resource (🔓)

GlobalModsManifestModelManifestedModFile

Represents a manifested mod in a global mods manifest SnippetTuning resource (🔓)

Class GlobalModsManifestModel

Namespace: LlamaLogic.Packages.Models.GlobalManifest

Assembly: LlamaLogic.Packages.dll

Represents the global mods manifest SnippetTuning resource (🔓)

```
public sealed class GlobalModsManifestModel : Model, IModel<GlobalModsManifestModel>,
IModel, IXmlSerializable
```

Inheritance

object ← Model ← GlobalModsManifestModel

Implements

IModel<GlobalModsManifestModel>, IModel, IXmlSerializable

Inherited Members

Model.Decode(ResourceType, ReadOnlyMemory<byte>) , Model.GetModelType(ResourceType) , Model.GetName(Stream) , Model.GetNameAsync(Stream, CancellationToken) , Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

This model allows agents to easily create, read, and update the global mods manifest, which itself is an index of the player's installed packs and manifested mod files easily acessible to script mods.

Properties

DisabledPacks

Gets the collection of pack codes that the launcher has specified as disabled

```
public Collection<string> DisabledPacks { get; }
```

Property Value

Collection<string>

InstalledPacks

Gets the collection of pack codes for this game installation's installed packs

```
public Collection<string> InstalledPacks { get; }
```

Property Value

Collection<string>

ManifestedModFiles

Gets the collection of manifested mod files in the player's Mods folder

```
public Collection<GlobalModsManifestModelManifestedModFile> ManifestedModFiles { get; }
```

Property Value

Collection<GlobalModsManifestModelManifestedModFile>

ResourceKey

Gets the ResourceKey always used by the global mods manifest

```
public static ResourceKey ResourceKey { get; }
```

Property Value

ResourceKey

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔄 🏃)

```
public static GlobalModsManifestModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

GlobalModsManifestModel

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

Class GlobalModsManifestModelManifestedModFile

Namespace: LlamaLogic.Packages.Models.GlobalManifest

Assembly: LlamaLogic.Packages.dll

Represents a manifested mod in a global mods manifest SnippetTuning resource (🔓)

```
public sealed class GlobalModsManifestModelManifestedModFile : IXmlSerializable
```

Inheritance

object ← GlobalModsManifestModelManifestedModFile

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Hashes

Gets the union of SubsumedHashes and the agent's calculated manifest hash for the mod file

```
public HashSet<ImmutableArray<byte>> Hashes { get; }
```

Property Value

HashSet<ImmutableArray<byte>>

ManifestKey

Gets/sets the ResourceKey of the mod manifest SnippetTuning of the mod file if it's a package

```
public ResourceKey? ManifestKey { get; set; }
```

Property Value

ResourceKey?

ManifestTuningName

Gets the tuning name of the mod manifest SnippetTuning of the mod file if it's a package

```
public string? ManifestTuningName { get; set; }
```

Property Value

string

ModsFolderPath

Gets/sets the relative path from the root of the Mods folder to the mod file

```
public string ModsFolderPath { get; set; }
```

Property Value

string

Namespace LlamaLogic.Packages.Models.ModFileManifest

Classes

ModFileManifestModel

Represents a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelRecommendedPack

Represents a recommended mod in a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelRepurposedLanguage

Represents a repurposed language in a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelRequiredMod

Represents a required mod in a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelTranslator

Represents the contribution of a translator in a mod file manifest SnippetTuning resource (🔓)

Class ModFileManifestModel

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModel : Model, IModel<ModFileManifestModel>,
IModel, IXmlSerializable
```

Inheritance

object ← Model ← ModFileManifestModel

Implements

IModel<ModFileManifestModel>, IModel, IXmlSerializable

Inherited Members

Model.Decode(ResourceType, ReadOnlyMemory<byte>) , Model.GetModelType(ResourceType) , Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object)

Remarks

This model allows callers to easily create, read, and update a mod file manifest. These manifests are a format sponsored by the Llama Logic team to permit creators to specify the dependency requirements of their mods.

Use in Mod Package (`.package`) files

When originally creating packages, place one and only one mod file manifest SnippetTuning resource in the package and use sufficiently unique values for its FullInstance and TuningName. If an external application encounters multiple SnippetTuning manifest resources in a package, it must assume that a manifest un-aware tool has merged multiple manifested packages and process each manifest accordingly.

Avoid using Group `0x00000000` as this is informally reserved for Maxis. It would be a bizarre miracle if they started incorporating dependency information using a community format. Regardless, let's not make them feel unwelcome and hold their spot for them.

Use in Script Mod Archive (`.ts4script`) files

Because these files are just ZIP archives with a different extension and their contents are not flatly merged by the game as the contents of package files are, there is no risk of collision. Also, since they need not contend with game resource management, they need not be in an arcane format like XML, so they are in YAML instead. The static parsing methods of this type expect YAML representations of manifests and the ToString() method produces them. Name your mod file manifest resource `llamalogic.modfilemanifest.yml` and put it in the root of your `.ts4script` archive.

Properties

ContactEmail

Gets/sets the email address which can be used to contact this mod's creator

```
[YamlMember(Order = 7, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? ContactEmail { get; set; }
```

Property Value

string

ContactUrl

Gets/sets the URL which can be used to contact this mod's creator

```
[YamlMember(Order = 8, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? ContactUrl { get; set; }
```

Property Value

Uri

Creators

Gets the names of the creators of the mod

```
[YamlMember(Order = 3, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Creators { get; }
```

Property Value

Collection <string>

Description

Gets/sets the description of the mod

```
[YamlMember(Order = 2, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? Description { get; set; }
```

Property Value

string

ElectronicArtsPromoCode

Gets/sets the promo code it is suggested the player use during check out in the EA Store if purchasing a pack for use with this mod

```
[YamlMember(Order = 21, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? ElectronicArtsPromoCode { get; set; }
```

Property Value

string

ExcludedEntries

Gets the names of entries in a script mod archive to be excluded from the Hash

```
[YamlMember(Order = 15, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public HashSet<string> ExcludedEntries { get; }
```

Property Value

HashSet<string>

Exclusivities

Gets the globally unique names of the exclusivities of this mod, causing it to be incompatible with other mods which share one or more of them

```
[YamlMember(Order = 18, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Exclusivities { get; }
```

Property Value

Collection<string>

Features

Gets the names of the features unique to this mod which it offers to other mods as a dependency

```
[YamlMember(Order = 17, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Features { get; }
```

Property Value

Collection<string>

FundingUrl

Gets/sets the URL to which players can go to offer support to the mod creator

```
[YamlMember(Order = 6, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? FundingUrl { get; set; }
```

Property Value

Uri

Hash

Gets/sets the hash of the mod file

```
[YamlMember(Order = 13, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public ImmutableArray<byte> Hash { get; set; }
```

Property Value

ImmutableArray<byte>

HashResourceKeys

Gets the resource keys for the resources included in the Hash

```
[YamlMember(Order = 14, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public HashSet<ResourceKey> HashResourceKeys { get; }
```

Property Value

HashSet<ResourceKey>

IncompatiblePacks

Gets the list of pack codes identifying the packs incompatible with this mod (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 22, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> IncompatiblePacks { get; }
```

Property Value

Collection<string>

MessageToTranslators

Gets/sets the message to translators by this mod's creator

```
[YamlMember(Order = 10, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? MessageToTranslators { get; set; }
```

Property Value

string

Name

Gets/sets the name of the mod

```
[YamlMember(Order = 1, DefaultValuesHandling = DefaultValuesHandling.OmitDefaults)]
public required string Name { get; set; }
```

Property Value

string

RecommendedPacks

Gets the list of pack codes identifying the packs required by this mod (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 20, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelRecommendedPack> RecommendedPacks { get; }
```

Property Value

Collection <ModFileManifestModelRecommendedPack>

RepurposedLanguages

Gets the list of languages repurposed by this mod

```
[YamlMember(Order = 12, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelRepurposedLanguage> RepurposedLanguages { get; }
```

Property Value

Collection<ModFileManifestModelRepurposedLanguage>

RequiredMods

Gets the list of mods required by this mod

```
[YamlMember(Order = 23, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelRequiredMod> RequiredMods { get; }
```

Property Value

Collection<ModFileManifestModelRequiredMod>

RequiredPacks

Gets the list of pack codes identifying the packs required by this mod (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 19, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> RequiredPacks { get; }
```

Property Value

Collection<string>

ResourceName

Gets the name of this resource if it has one

```
[YamlIgnore]
public override string? ResourceName { get; }
```

Property Value

string

SubsumedHashes

Gets the hashes of previous versions of this mod in for which I stand even though my hash is different

```
[YamlMember(Order = 16, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public HashSet<ImmutableArray<byte>> SubsumedHashes { get; }
```

Property Value

HashSet <ImmutableArray <byte > >

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet <ResourceType>

TranslationSubmissionUrl

Gets/sets the URL at which translators may submit translations to this mod's creator

```
[YamlMember(Order = 11, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? TranslationSubmissionUrl { get; set; }
```

Property Value

Uri

Translators

Gets the list of translators who have contributed translations to this mod

```
[YamlMember(Order = 9, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelTranslator> Translators { get; }
```

Property Value

Collection<ModFileManifestModelTranslator>

TuningFullInstance

Gets the SnippetTuning decimal conversion of the FullInstance for the mod file manifest

```
[YamlMember(Order = -2, DefaultValuesHandling = DefaultValuesHandling.OmitDefaults)]
public ulong TuningFullInstance { get; set; }
```

Property Value

ulong

TuningName

Gets the SnippetTuning name for the mod file manifest

```
[YamlMember(Order = -1, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? TuningName { get; set; }
```

Property Value

string

Url

Gets/sets the URL to which players can go to find more information about this mod

```
[YamlMember(Order = 5, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? Url { get; set; }
```

Property Value

Uri

Version

Gets/sets the version of this mod

```
[YamlMember(Order = 4, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? Version { get; set; }
```

Property Value

string

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔁 🏃)

```
public static ModFileManifestModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

ModFileManifestModel

DeleteModFileManifests(ZipFile)

Deletes all mod file manifest resources in the specified `scriptMod`

```
public static void DeleteModFileManifests(ZipFile scriptMod)
```

Parameters

`scriptMod` ZipFile

DeleteModFileManifests(DataBasePackedFile)

Deletes all mod file manifest resources in the specified `package`

```
public static void DeleteModFileManifests(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

DeleteModFileManifestsAsync(DataBasePackedFile)

Deletes all mod file manifest resources in the specified `package`, asynchronously

```
public static Task DeleteModFileManifestsAsync(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

Task

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

GetFileSha256Hash(string)

Gets the SHA 256 hash of the content of the file located at `filePath`

```
public static ImmutableArray<byte> GetFileSha256Hash(string filePath)
```

Parameters

`filePath` string

Returns

ImmutableArray<byte>

GetFileSha256HashAsync(string, CancellationToken)

Gets the SHA 256 hash of the content of the file located at `filePath` asynchronously

```
public static Task<ImmutableArray<byte>> GetFileSha256HashAsync(string filePath,
CancellationToken cancellationToken = default)
```

Parameters

`filePath` string

`cancellationToken` CancellationToken

Returns

Task<ImmutableArray<byte>>

GetModFileHash(ZipFile, HashSet<string>)

Gets the hash for the specified `scriptMod`, excluding entries with the names specified by `excludedEntries`

```
public static ImmutableArray<byte> GetModFileHash(ZipFile scriptMod,
HashSet<string> excludedEntries)
```

Parameters

`scriptMod` ZipFile

`excludedEntries` HashSet<string>

Returns

ImmutableArray<byte>

GetModFileHash(DataBasePackedFile, HashSet<ResourceKey>)

Gets the hash for the specified `package` using the specified `hashResourceKeys`

```
public static ImmutableArray<byte> GetModFileHash(DataBasePackedFile package,
HashSet<ResourceKey> hashResourceKeys)
```

Parameters

`package` DataBasePackedFile

`hashResourceKeys` HashSet<ResourceKey>

Returns

ImmutableArray<byte>

GetModFileHashAsync(DataBasePackedFile, HashSet<ResourceKey>, CancellationToken)

Gets the hash for the specified `package` using the specified `hashResourceKeys`, asynchronously

```
public static Task<ImmutableArray<byte>> GetModFileHashAsync(DataBasePackedFile package,
HashSet<ResourceKey> hashResourceKeys, CancellationToken cancellationToken = default)
```

Parameters

`package` DataBasePackedFile

`hashResourceKeys` HashSet<ResourceKey>

`cancellationToken` CancellationToken

Returns

Task<ImmutableArray<byte>>

GetModFileManifest(ZipFile)

Gets the mod file manifest for the specified `scriptMod`, if it has one

```
public static ModFileManifestModel? GetModFileManifest(ZipFile scriptMod)
```

Parameters

`scriptMod` ZipFile

Returns

ModFileManifestModel

GetModFileManifest(DataBasePackedFile)

Gets the mod file manifest for the specified `package`, if it has one

```
public static ModFileManifestModel? GetModFileManifest(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

ModFileManifestModel

GetModFileManifestAndKey(DataBasePackedFile)

Gets the mod file manifest for the specified `package` and its ResourceKey, if the `package` has one

```
public static (ResourceKey, ModFileManifestModel?)
GetModFileManifestAndKey(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

(ResourceKey, ModFileManifestModel)

GetModFileManifestAndKeyAsync(DataBasePackedFile)

Gets the mod file manifest for the specified `package`, asynchronously, if the `package` has one

```
public static Task<(ResourceKey, ModFileManifestModel?)>
GetModFileManifestAndKeyAsync(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

Task<(ResourceKey, ModFileManifestModel)>

GetModFileManifestAsync(ZipFile)

Gets the mod file manifest for the specified `scriptMod`, asynchronously, if it has one

```
public static Task<ModFileManifestModel?> GetModFileManifestAsync(ZipFile scriptMod)
```

Parameters

`scriptMod` ZipFile

Returns

Task<ModFileManifestModel>

GetModFileManifestAsync(DataBasePackedFile)

Gets the mod file manifest for the specified `package`, asynchronously, if it has one

```
public static Task<ModFileManifestModel?> GetModFileManifestAsync(DataBasePackedFile
package)
```

Parameters

`package` DataBasePackedFile

Returns

Task<ModFileManifestModel>

GetModFileManifests(DataBasePackedFile)

Gets the mod file manifests for the specified `package` (for use by editors checking for packages which may have been merged by manifest unaware tooling)

```
public static IReadOnlyDictionary<ResourceKey, ModFileManifestModel>
GetModFileManifests(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

IReadOnlyDictionary<ResourceKey, ModFileManifestModel>

GetModFileManifestsAsync(DataBasePackedFile)

Gets the mod file manifests for the specified `package`, asynchronously (for use by editors checking for packages which may have been merged by manifest unaware tooling)

```
public static Task<IReadOnlyDictionary<ResourceKey, ModFileManifestModel>>
GetModFileManifestsAsync(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

Task<IReadOnlyDictionary<ResourceKey, ModFileManifestModel>>

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Parse(string)

Parses a string into a ModFileManifestModel

```
public static ModFileManifestModel Parse(string s)
```

Parameters

`s` string

The string to parse

Returns

ModFileManifestModel

Exceptions

FormatException

`s` is not in the correct format

Parse(string, IFormatProvider?)

Parses a string into a ModFileManifestModel

```
public static ModFileManifestModel Parse(string s, IFormatProvider? provider)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

Returns

ModFileManifestModel

Exceptions

FormatException

`s` is not in the correct format

SetModFileManifest(ZipFile, ModFileManifestModel)

Set the specified `manifest` for the specified `scriptMod`

```
public static void SetModFileManifest(ZipFile scriptMod, ModFileManifestModel manifest)
```

Parameters

`scriptMod` ZipFile

`manifest` ModFileManifestModel

SetModFileManifest(DataBasePackedFile, ModFileManifestModel)

Set the specified `manifest` for the specified `package`

```
public static void SetModFileManifest(DataBasePackedFile package,
ModFileManifestModel manifest)
```

Parameters

`package` DataBasePackedFile

`manifest` ModFileManifestModel

SetModFileManifestAsync(ZipFile, ModFileManifestModel)

Set the specified `manifest` for the specified `scriptMod`, asynchronously

```
public static Task SetModFileManifestAsync(ZipFile scriptMod, ModFileManifestModel manifest)
```

Parameters

`scriptMod` ZipFile

`manifest` ModFileManifestModel

Returns

Task

SetModFileManifestAsync(DataBasePackedFile, ModFileManifestModel)

Set the specified `manifest` for the specified `package`, asynchronously

```
public static Task SetModFileManifestAsync(DataBasePackedFile package,
ModFileManifestModel manifest)
```

Parameters

`package` DataBasePackedFile

`manifest` ModFileManifestModel

Returns

Task

ToString()

Generates the YAML representation of the ModFileManifestModel

```
public override string ToString()
```

Returns

string

TryParse(string?, out ModFileManifestModel)

Tries to parse a string into a ModFileManifestModel

```
public static bool TryParse(string? s, out ModFileManifestModel result)
```

Parameters

`s` string

The string to parse

`result` ModFileManifestModel

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

TryParse(string?, IFormatProvider?, out ModFileManifestModel)

Tries to parse a string into a ResourceKey

```
public static bool TryParse(string? s, IFormatProvider? provider, out
ModFileManifestModel result)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

`result` ModFileManifestModel

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

Class ModFileManifestModelRecommendedPack

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a recommended mod in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelRecommendedPack : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelRecommendedPack

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

PackCode

Gets/sets the pack code indicating the pack that is recommended (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 1)]
public string PackCode { get; set; }
```

Property Value

string

Reason

Gets/sets the reason for the pack recommendation

```
[YamlMember(Order = 2)]
public string Reason { get; set; }
```

Property Value

string

Class ModFileManifestModelRepurposedLanguage

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a repurposed language in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelRepurposedLanguage : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelRepurposedLanguage

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ActualLocale

Gets/sets the IETF BCP 47 language tag identifying the non-game-supported language this mod actually contains in the domain ordinarily reserved to GameLocale

```
[YamlMember(Order = 2)]
public string ActualLocale { get; set; }
```

Property Value

string

GameLocale

Gets/sets the IETF BCP 47 language tag identifying the game-supported language this mod is repurposing

```
[YamlMember(Order = 1)]
public string GameLocale { get; set; }
```

Property Value

string

Remarks

The IETF BCP 47 language tags currently supported by The Sims 4 at time of writing are:

Tag	Description
en-US	American English | English (United States) | Full instance prefix: `0x00`
zh-CN	简体中文 | Chinese (Simplified, Mainland China) | Full instance prefix: `0x01`
zh-TW	繁體中文 | Chinese (Traditional, Taiwan) | Full instance prefix: `0x02`
cz	Čeština | Czech | Full instance prefix: `0x03`
da	Dansk | Danish | Full instance prefix: `0x04`
nl	Nederlands | Dutch | Full instance prefix: `0x05`
fi	Suomi | Finnish | Full instance prefix: `0x06`
fr-FR	Français | French (France) | Full instance prefix: `0x07`
de	Deutsch | German | Full instance prefix: `0x08`
it	Italiano | Italian | Full instance prefix: `0x0b`
ja	日本語 | Japanese | Full instance prefix: `0x0c`
ko	한국어 | Korean | Full instance prefix: `0x0d`
nb	Norsk | Norwegian | Full instance prefix: `0x0e`
pl	Polski | Polish | Full instance prefix: `0x0f`

Tag	Description
pt-BR	Português | Portuguese (Brazil) | Full instance prefix: `0x11`
ru	Русский | Russian | Full instance prefix: `0x12`
es-ES	Español | Spanish (Spain) | Full instance prefix: `0x13`
sv	Svenska | Swedish | Full instance prefix: `0x15`

Class ModFileManifestModelRequiredMod

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a required mod in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelRequiredMod : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelRequiredMod

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Creators

Gets the names of the creators of the dependency mod

```
[YamlMember(Order = 2, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Creators { get; }
```

Property Value

Collection<string>

Hashes

Gets the hashes of the mod files that must be present in order for this dependency to be fulfilled

```
[YamlMember(Order = 7)]
public HashSet<ImmutableArray<byte>> Hashes { get; }
```

Property Value

HashSet <ImmutableArray <byte> >

IgnoreIfHashAvailable

Instructs agents to ignore fulfilling this dependency if this hash is present in the player's catalog of mods

```
[YamlMember(Order = 8, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public ImmutableArray<byte> IgnoreIfHashAvailable { get; set; }
```

Property Value

ImmutableArray <byte>

IgnoreIfHashUnavailable

Instructs agents to ignore fulfilling this dependency if this hash is not present in the player's catalog of mods

```
[YamlMember(Order = 9, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public ImmutableArray<byte> IgnoreIfHashUnavailable { get; set; }
```

Property Value

ImmutableArray <byte>

IgnoreIfPackAvailable

Instructs agents to ignore fulfilling this dependency if the pack identified by this pack code is present (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 10, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? IgnoreIfPackAvailable { get; set; }
```

Property Value

string

IgnoreIfPackUnavailable

Instructs agents to ignore fulfilling this dependency if the pack identified by this pack code is not present (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 11, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? IgnoreIfPackUnavailable { get; set; }
```

Property Value

string

Name

Gets/sets the name of the dependency mod

```
[YamlMember(Order = 1)]
public string Name { get; set; }
```

Property Value

string

RequiredFeatures

Gets the names of the features of the dependency mod which the dependent mod requires

```
[YamlMember(Order = 5, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> RequiredFeatures { get; }
```

Property Value

Collection<string>

RequirementIdentifier

Gets/sets an identifier that, when shared with a group of other dependency mods for a single dependent, indicates that only one member of the group need be present for the dependent to be satisfied

```
[YamlMember(Order = 6, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? RequirementIdentifier { get; set; }
```

Property Value

string

Url

Gets/sets the URL to which players can go to find more information about this dependency mod

```
[YamlMember(Order = 4, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? Url { get; set; }
```

Property Value

Uri

Version

Gets/sets the version of this dependency mod

```
[YamlMember(Order = 3, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? Version { get; set; }
```

Property Value

string

Class ModFileManifestModelTranslator

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents the contribution of a translator in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelTranslator : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelTranslator

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Culture

Gets/sets the value of the Language property using a CultureInfo object

```
[YamlIgnore]
public CultureInfo? Culture { get; set; }
```

Property Value

CultureInfo

Language

Gets/sets the IETF BCP 47 language tag identifying the language to which this translator contributed

```
[YamlMember(Order = 2)]
public string Language { get; set; }
```

Property Value

string

Name

Gets/sets the name of the translator

```
[YamlMember(Order = 1)]
public string Name { get; set; }
```

Property Value

string

Namespace LlamaLogic.Packages.Models.SimData

Classes

SimDataColumn
: Represents a column of a SimDataSchema (🔒)

SimDataInstance
: Represents a named, top-level SimDataObject in a SimDataModel (🔓)

SimDataModel
: Represents a SimData resource (🔓)

SimDataObject
: Represents an object in a SimData resource: a row of values described by a SimDataSchema (🔓)

SimDataSchema
: Represents the schema of a SimDataObject (🔒)

SimDataVariant
: Represents a VARIANT value: a type hash and an optional value of any SimDataType (🔒)

SimDataVector
: Represents a VECTOR value: a sequence of elements of one SimDataType (🔓)

Enums

SimDataType
: A type of data within a SimData resource

SimDataVersion
: Known versions of the format for SimData resources

Class SimDataColumn

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents a column of a SimDataSchema (🔒)

```
public sealed class SimDataColumn
```

Inheritance

object ← SimDataColumn

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SimDataColumn(string, SimDataType, ushort)

Initializes a new SimDataColumn

```
public SimDataColumn(string name, SimDataType type, ushort flags = 0)
```

Parameters

`name` string

The name of the column

`type` SimDataType

The type of the column's values

`flags` ushort

The column's flags (Maxis sets none)

Exceptions

ArgumentOutOfRangeException

`type` is not a defined SimDataType

Properties

Flags

Gets the column's flags

```
public ushort Flags { get; }
```

Property Value

ushort

Name

Gets the name of the column

```
public string Name { get; }
```

Property Value

string

Type

Gets the type of the column's values

```
public SimDataType Type { get; }
```

Property Value

SimDataType

Class SimDataInstance

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents a named, top-level SimDataObject in a SimDataModel (🔓)

```
public sealed class SimDataInstance
```

Inheritance

object ← SimDataInstance

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SimDataInstance(string, SimDataObject)

Initializes a new SimDataInstance

```
public SimDataInstance(string name, SimDataObject @object)
```

Parameters

`name` string

`object` SimDataObject

Properties

Name

Gets/sets the name of the instance

```
public string Name { get; set; }
```

Property Value

string

Object

Gets the object

```
public SimDataObject Object { get; }
```

Property Value

SimDataObject

Class SimDataModel

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents a SimData resource (🔓)

```
public sealed class SimDataModel : Model, IModel<SimDataModel>, IModel
```

Inheritance

object ← Model ← SimDataModel

Implements

IModel<SimDataModel>, IModel

Inherited Members

Model.Decode(ResourceType, ReadOnlyMemory<byte>) , Model.GetModelType(ResourceType) , Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

A SimData resource is a set of named Instances, each an object described by a SimDataSchema whose values may refer to further objects, strings, vectors and variants. Decode(ReadOnlyMemory<byte>) reads any well-formed layout, including those written by community tools, and reads a reference whose offset is zero (which could only refer to itself, and which some community tools write) as null. It rejects a resource in which a row is referred to more than once, which Maxis never writes and which the model, being a tree, cannot represent. Encode() always lays the resource out the way Maxis does, so decoding and encoding a resource Maxis wrote reproduces it byte for byte.

Properties

Instances

Gets the named, top-level objects of the resource

```
public Collection<SimDataInstance> Instances { get; }
```

Property Value

Collection<SimDataInstance>

Pack

Gets/sets the pack to which the resource belongs, which is the `mUnused` field of the resource's header (ignored for SmartSimOriginalEdition)

```
public uint Pack { get; set; }
```

Property Value

uint

Remarks

The value is that of the game's `sims4.common.Pack` enumeration, such as `0` for the base game and `3` for EP01.

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

Schemas

Gets the schemas of the objects in the resource, in the order Encode() writes them

```
public IReadOnlyList<SimDataSchema> Schemas { get; }
```

Property Value

IReadOnlyList<SimDataSchema>

Exceptions

InvalidOperationException

Two different schemas have the same hash, or an object contains itself

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Version

Gets/sets the version of the format of the resource

```
public SimDataVersion Version { get; set; }
```

Property Value

SimDataVersion

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔄 🏃)

```
public static SimDataModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

SimDataModel

Exceptions

InvalidDataException

The data is not a well-formed SimData resource

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

Exceptions

InvalidOperationException

The resource cannot be encoded: Version is not supported or does not support a value used, two different schemas have the same hash, two instances of the same schema have different names, or an object or vector contains itself

FromXml(XDocument)

Creates a SimDataModel from the XML representation of a SimData resource that Sims 4 Studio exports and imports

```
public static SimDataModel FromXml(XDocument document)
```

Parameters

`document` XDocument

Returns

SimDataModel

Remarks

Fields are matched to columns by exact name, and every column of an object must have exactly one field. A string field written as an empty element (`<T name="..." />`) is read as null. Type names written by Sims4Toolkit (`Int8` and `UInt8`) are accepted along with Sims 4 Studio's. A schema without a `schema_hash` attribute gets its ComputedHash.

Exceptions

InvalidDataException

> `document` is not a valid representation of a SimData resource

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

ToXml()

Gets the XML representation of the resource that Sims 4 Studio exports and imports

```
public XDocument ToXml()
```

Returns

XDocument

Remarks

A null string is written as an empty element (`<T name="..." />`) and an empty string as an element with empty content (`<T name="..."></T>`), which FromXml(XDocument) tells apart (Sims 4 Studio reads both as empty strings). An empty vector is written the same way as a null one, which Encode() also writes the same way.

Exceptions

InvalidOperationException

> Two different schemas have the same hash or the same name, a name or string contains characters XML cannot represent, or an object or vector contains itself

Class SimDataObject

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents an object in a SimData resource: a row of values described by a SimDataSchema (🔓)

```
public sealed class SimDataObject
```

Inheritance

object ← SimDataObject

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Each value's .NET type is determined by its column's SimDataType: BOOL is bool; CHAR8 and UINT8 are byte; INT8 is sbyte; INT16 is short; UINT16 is ushort; INT32 is int; UINT32 and LOCKEY are uint; INT64 is long; UINT64 and TABLESETREFERENCE are ulong; FLOAT is float; FLOAT2, FLOAT3 and FLOAT4 are Vector2, Vector3 and Vector4; RESOURCEKEY is ResourceKey; STRING8 and HASHEDSTRING8 are string (or null; the hash of a HASHEDSTRING8 is computed when encoding); OBJECT is SimDataObject, VECTOR is SimDataVector, and VARIANT is SimDataVariant (each or null).

Constructors

SimDataObject(SimDataSchema)

Initializes a new SimDataObject with the specified `schema`, all of its values defaulted (zero, or null for references)

```
public SimDataObject(SimDataSchema schema)
```

Parameters

`schema` SimDataSchema

Properties

this[int]

Gets/sets the value of the column at the specified `columnIndex`

```
public object? this[int columnIndex] { get; set; }
```

Parameters

`columnIndex` int

Property Value

object

Exceptions

ArgumentException

The value is not of the .NET type the column's SimDataType requires

this[string]

Gets/sets the value of the column with the specified `columnName`

```
public object? this[string columnName] { get; set; }
```

Parameters

`columnName` string

Property Value

object

Exceptions

ArgumentException

There is no such column, or the value is not of the .NET type the column's SimDataType requires

Schema

Gets the schema of the object

```
public SimDataSchema Schema { get; }
```

Property Value

SimDataSchema

Class SimDataSchema

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents the schema of a SimDataObject (🔒)

```
public sealed class SimDataSchema
```

Inheritance

object ← SimDataSchema

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

The order of Columns is the order in which the columns' values are laid out in each row. For schemas the game reads with a fixed layout (e.g. `Client_` and `Renderer.` schemas), this order must match the game's, which is why Decode(ReadOnlyMemory<byte>) preserves it. The game derives a schema's hash from its name and layout (see ComputedHash), so changing the columns or their order changes the hash the game expects.

Constructors

SimDataSchema(string, IEnumerable<SimDataColumn>)

Initializes a new SimDataSchema whose Hash is its ComputedHash

```
public SimDataSchema(string name, IEnumerable<SimDataColumn> columns)
```

Parameters

`name` string

The name of the schema

`columns` IEnumerable<SimDataColumn>

The columns of the schema, in layout order

Exceptions

ArgumentException

`columns` is empty, contains null, or contains two columns with the same name

SimDataSchema(string, uint, IEnumerable<SimDataColumn>)

Initializes a new SimDataSchema

```
public SimDataSchema(string name, uint hash, IEnumerable<SimDataColumn> columns)
```

Parameters

`name` string

The name of the schema

`hash` uint

The hash identifying the schema

`columns` IEnumerable<SimDataColumn>

The columns of the schema, in layout order

Exceptions

ArgumentException

`columns` is empty, contains null, or contains two columns with the same name

Properties

Columns

Gets the columns of the schema, in layout order

```
public IReadOnlyList<SimDataColumn> Columns { get; }
```

Property Value

IReadOnlyList<SimDataColumn>

ComputedHash

Gets the hash the game derives from the schema's Name and the layout of its Columns

```
public uint ComputedHash { get; }
```

Property Value

uint

Remarks

The game computes this as the sum of the Fnv32 hash of the lowercased name, the row size shifted left 16 bits, the number of columns, and, for each column, the Fnv32 hash of its lowercased name XOR its type shifted left 16 bits XOR its offset in the row shifted left 8 bits. Column flags do not contribute.

Hash

Gets the hash identifying the schema

```
public uint Hash { get; }
```

Property Value

uint

Name

Gets the name of the schema

```
public string Name { get; }
```

Property Value

string

Methods

GetColumnIndex(string)

Gets the index of the column with the specified `name`, or null if there is no such column

```
public int? GetColumnIndex(string name)
```

Parameters

`name` string

Returns

int?

Enum SimDataType

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

A type of data within a SimData resource

```
public enum SimDataType
```

Fields

`BOOL = 0`

A bool expressed with an entire byte (`0` is false, any other value is true)

`CHAR8 = 1`

A single byte of a UTF-8 string

`FLOAT = 10`

A float

`FLOAT2 = 15`

You might think it's a double, but nope: two floats

`FLOAT3 = 16`

An obvious trend is developing, for this is three floats

`FLOAT4 = 17`

(You know when, every once in a while, there's a great trilogy and someone messes it up?) Here are your four floats

`HASHEDSTRING8 = 12`

An int offset to a null-terminated UTF-8 string and its accompanying Fnv32 hash

`INT16 = 4`

A short

`INT32 = 6`

An int

`INT64 = 8`

A long

`INT8 = 2`

An sbyte

`LOCKEY = 20`

A four-byte key used to uniquely identify localized strings in The Sims 4 (typically the Fnv32 hash of the string in its original language)

`OBJECT = 13`

An int offset to an object

`RESOURCEKEY = 19`

A ResourceKey laid out by FullInstance, then Type, then Group

`STRING8 = 11`

An int offset to a null-terminated UTF-8 string

`TABLESETREFERENCE = 18`

It's not just about good manners from "How to Serve and Not Be Served", it's really about this ulong

Seriously, though, why have these when we already have UINT64, you ask?

> you're right that table set refs are just uint64s, but it is important to distinguish them because they mean different things. int64s are just literal numbers, table set refs are tuning IDs. so, when the game loads these, it knows that uint64 is just a number, so leave it alone, whereas a table set ref is tuning, so it needs to look that tuning up and use it as the value there rather than the number itself

— frankk, August 16, 2024 in Lot 51's Discord

`UINT16 = 5`

A ushort

`UINT32 = 7`

A uint

`UINT64 = 9`

A ulong

`UINT8 = 3`

A byte

`VARIANT = 21`

An int offset to an object accompanied by a uint type hash

`VECTOR = 14`

An int offset to an array of objects and a uint that is its size

Class SimDataVariant

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents a VARIANT value: a type hash and an optional value of any SimDataType (🔒)

```
public sealed class SimDataVariant
```

Inheritance

object ← SimDataVariant

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SimDataVariant(uint)

Initializes a new SimDataVariant with the specified `typeHash` and no value

```
public SimDataVariant(uint typeHash)
```

Parameters

`typeHash` uint

SimDataVariant(uint, SimDataType, object?)

Initializes a new SimDataVariant with the specified `typeHash` and a `value` of the specified `type`

```
public SimDataVariant(uint typeHash, SimDataType type, object? value)
```

Parameters

`typeHash` uint

The type hash of the variant

`type` SimDataType

The type of `value`

`value` object

The value, which may be null only for STRING8, HASHEDSTRING8, VECTOR and VARIANT

Exceptions

ArgumentOutOfRangeException

`type` is not a defined SimDataType

ArgumentException

`value` is not of the .NET type SimDataObject describes for `type`, or is null for OBJECT

Properties

Type

Gets the type of Value, or null if the variant has no value

```
public SimDataType? Type { get; }
```

Property Value

SimDataType?

TypeHash

Gets the type hash of the variant

```
public uint TypeHash { get; }
```

Property Value

uint

Value

Gets the value of the variant (null when Type is null, and possibly when it is a type that admits null)

```
public object? Value { get; }
```

Property Value

object

Class SimDataVector

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Represents a VECTOR value: a sequence of elements of one SimDataType (🔓)

```
public sealed class SimDataVector : Collection<object?>, IList<object?>, ICollection<object?
>, IReadOnlyList<object?>, IReadOnlyCollection<object?>, IEnumerable<object?>, IList,
ICollection, IEnumerable
```

Inheritance

object ← Collection<object> ← SimDataVector

Implements

IList<object>, ICollection<object>, IReadOnlyList<object>, IReadOnlyCollection<object>, IEnumerable<object>, IList, ICollection, IEnumerable

Inherited Members

Collection<object>.Add(object) , Collection<object>.Clear() , Collection<object>.Contains(object) , Collection<object>.CopyTo(object[], int) , Collection<object>.GetEnumerator() , Collection<object>.IndexOf(object) , Collection<object>.Insert(int, object) , Collection<object>.Remove(object) , Collection<object>.RemoveAt(int) , Collection<object>.Count , Collection<object>.this[int] , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Elements are of the .NET type SimDataObject describes for ElementType, except that elements of a vector of OBJECT cannot be null and must all have the same SimDataSchema.

Constructors

SimDataVector(SimDataType, IEnumerable<object?>?)

Initializes a new SimDataVector with the specified `elementType` and optional initial `elements`

```
public SimDataVector(SimDataType elementType, IEnumerable<object?>? elements = null)
```

Parameters

`elementType` SimDataType

`elements` IEnumerable<object>

Exceptions

ArgumentOutOfRangeException

`elementType` is not a defined SimDataType

ArgumentException

An element is not valid for `elementType`

Properties

ElementType

Gets the type of the vector's elements

```
public SimDataType ElementType { get; }
```

Property Value

SimDataType

Methods

InsertItem(int, object?)

Inserts an element into the Collection<T> at the specified index.

```
protected override void InsertItem(int index, object? item)
```

Parameters

`index` int

The zero-based index at which `item` should be inserted.

`item` object

The object to insert. The value can be null for reference types.

Exceptions

ArgumentOutOfRangeException

`index` is less than zero.

-or-

`index` is greater than Count.

SetItem(int, object?)

Replaces the element at the specified index.

```
protected override void SetItem(int index, object? item)
```

Parameters

`index` int

The zero-based index of the element to replace.

`item` object

The new value for the element at the specified index. The value can be null for reference types.

Exceptions

ArgumentOutOfRangeException

`index` is less than zero.

-or-

`index` is greater than Count.

Enum SimDataVersion

Namespace: LlamaLogic.Packages.Models.SimData

Assembly: LlamaLogic.Packages.dll

Known versions of the format for SimData resources

```
[SuppressMessage("Design", "CA1008: Enums should have zero value", Justification = "We do
not know what the proper name of such a value would be.")]
[SuppressMessage("Design", "CA1028: Enum Storage should be Int32", Justification = "This
is determined by the binary format created by Maxis. So sorry to clearly agitate you
so much.")]
public enum SimDataVersion : uint
```

Fields

`SmartSimOriginalEdition = 256`

The edition of the format which shipped with the original release of The Sims 4 on September 2, 2014

`VariantSupport = 257`

The edition of the format which shipped with The Sims 4 (v1.5.139) on February 17, 2015, introducing support for VARIANT and allowing the creation of CombinedTuning resources